Chapter 1

Equivariant Cohomology

In the introduction we said that equivariant cohomology is cohomology that remembers a group action. How do we build it?

A naive idea is to construct a space from the original space X together with another specific space Y which is somehow derived from the group G. So, roughly speaking, the ordinary cohomology of this new space captures the equivariant information.

This idea is realised in the Borel construction, named after Borel [13]. It is convenient for computations, and it also has an interpretation in terms of quotient stacks, which we sketch in Section 1.1.

Our language is that of symplectic geometry and algebraic topology. We say little about algebraic geometry and representation theory.

Throughout the thesis, we assume that G is a compact Lie group acting on a paracompact Hausdorff space. This includes zero-dimensional compact Lie groups, such as $\mathbb{Z}/p$ equipped with the discrete topology.

1.1 The Borel construction

Definition 1.1.1. Let G be a topological group. A principal G-bundle $EG \longrightarrow BG$ is called a *universal principal G-bundle* if EG is contractible and the G-action on EG is free. The quotient $BG := EG/G$ is called the *classifying space* of G.

Here we choose a contractible space EG on which G acts freely, so that $EG \simeq \mathrm{pt}$, allowing us to form a quotient in the homotopy-theoretic sense.

Definition 1.1.2. Let G be a topological group acting on a space X on the left. The *Borel construction* of X is $X_G := EG \times_G X := (EG \times X)/G$, where we identify $(pg, x) \sim (p, gx)$. The equivariant cohomology of X is then defined by $H^*_G(X) := H^*(X_G)$.

A geometric interpretation is illustrated by the following diagram.

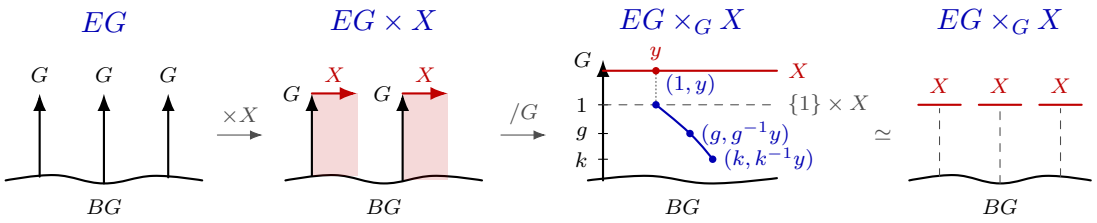


Figure 1.1: The Borel construction $EG \times_G X$ as an X-bundle over BG.

Tracing each orbit back to a chosen slice, here we choose $\{1\} \times X$, gives a schematic picture of the Borel construction. In this way, the information carried by the G-action is retained very naively.

The existence of EG is provided by Milnor's construction using the infinite topological join [57], which can in fact be shown to be contractible using Whitehead's theorem. We briefly explain why BG is called the *classifying space*; this is summarised by the following proposition.

Proposition 1.1.3 ([54, Chapter 23, §8])**.** *Let X be a paracompact Hausdorff space. There is a natural bijection*

$$[X, BG] \cong \left\{ \begin{array}{c} \textit{isomorphism classes of principal} \\ G\textit{-bundles over } X \end{array} \right\}.$$

Under this correspondence, a map $f\colon X \to BG$ determines the pullback bundle $f^*EG \to X$.

Example 1.1.4. Let $G = S^1$. We claim that we can take

$$ES^1 = S^\infty := \bigcup_{n \geq 0} S^{2n+1}.$$

Here S^1 acts freely on S^{2n+1}, the unit vectors in $\mathbb{R}^{2(n+1)} = \mathbb{C}^{n+1}$, by rotation $z \mapsto \lambda z$, and S^∞ is well known to be contractible. So

$$BS^1 = ES^1/S^1 = \bigcup_{n \geq 0} S^{2n+1}/S^1 = \bigcup_{n \geq 0} \mathbb{CP}^n =: \mathbb{CP}^\infty.$$

Example 1.1.5. Let $G = \mathbb{Z}/p$. Viewing $\mathbb{Z}/p$ as the group of p-th roots of unity, it acts freely on $S^\infty \subset \mathbb{C}^\infty$ by scalar multiplication. Thus again we may take $E(\mathbb{Z}/p) = S^\infty$, and hence $B(\mathbb{Z}/p) = S^\infty/(\mathbb{Z}/p) = L^\infty(p)$.

In particular, when $p = 2$, the action is the antipodal action and $B(\mathbb{Z}/2) = \mathbb{R}P^\infty$.

When G is discrete, for example finite, BG is a $K(G,1)$, and $H^*(BG;R)$ agrees with the group cohomology $H^*(G;R)$ [17]. This is why the cohomology of BG is also commonly referred to as group cohomology.

The following proposition is the heart of the Borel construction.

Proposition 1.1.6. *Let $q : EG \times_G X \to X/G$ be the natural map $q([e,x]) = [x]$. Fix $x_0 \in X$, and write $H = G_{x_0} = \{g \in G : gx_0 = x_0\}$. Then*

$$q^{-1}([x_0]) \cong EG \times_G (G \cdot x_0) \cong EG/H \simeq BH.$$

Proof. The point $[x_0] \in X/G$ represents the whole orbit $G \cdot x_0$, so $q^{-1}([x_0]) = \{[e,t] : t \in G \cdot x_0\} \cong EG \times_G (G \cdot x_0)$.

By the orbit-stabiliser identification, $G \cdot x_0 \cong G/H$. Thus it remains to identify $EG \times_G (G/H)$ with EG/H. The map is $\Phi([e,gH]) = egH$, with inverse $eH \mapsto [e,H]$. One checks directly that these maps are well-defined and inverse to each other. Finally, by restricting the G-action on EG to H, the space EG is a free contractible H-space, hence a model for EH. Therefore $EG/H \simeq BH$. □

For readers who know stacks, this suggests that the Borel construction gives the homotopy type of the topological quotient stack $[X/G]$. Here is a schematic picture.

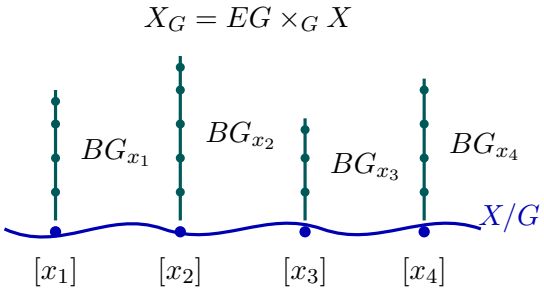


Figure 1.2: X_G is X/G with BG_x inserted over each $[x]$, only as a picture.

More precisely, X_G is a classifying space of the action groupoid $G \ltimes X$ [62]. We will not go further into this here, but the reader should keep Figure 1.2 in mind.

Immediately, we have the following corollaries from the picture.

Corollary 1.1.7. *If G acts freely on X, then $X_G \simeq X/G$, that is, $[X/G] \simeq X/G$. Hence*

$$H^*_G(X;R) \cong H^*(X/G;R)$$

for any coefficient ring R, and in fact in any cohomology theory.

Corollary 1.1.8. *If $x \in X$ has stabiliser $H = G_x$, then $G \cdot x \cong G/H$, and*

$$H^*_G(G \cdot x; R) \cong H^*_G(G/H; R) \cong H^*(BH; R).$$

Thus an orbit with stabiliser H behaves, in Borel equivariant cohomology, like the classifying space BH. A free orbit contributes $H^*(\mathrm{pt}; R)$, while a fixed point contributes $H^*(BG; R)$.

Example 1.1.9. Let S^1 act on S^2 by weight $k > 0$, i.e. $\lambda \cdot z = \lambda^k z$. Now we compute $H^*_{S^1}(S^2; \mathbb{Z})$.

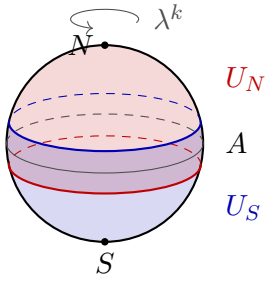


Figure 1.3: The cover $S^2 = U_N \cup U_S$, with $A = U_N \cap U_S$.

Consider $S^2 = U_N \cup U_S$ as in Figure 1.3, and apply the equivariant Mayer–Vietoris sequence:

$$\cdots \to H^{q-1}_{S^1}(A) \xrightarrow{\delta} H^q_{S^1}(S^2) \longrightarrow H^q_{S^1}(U_N) \oplus H^q_{S^1}(U_S) \longrightarrow H^q_{S^1}(A) \xrightarrow{\delta} \cdots$$

Clearly $H^*_{S^1}(U_N) \cong H^*_{S^1}(\mathrm{pt}) = H^*(BS^1) = \mathbb{Z}[u]$, $\deg u = 2$, and the same holds for U_S. Take $x \in A$. Rotating it gives a circle, and A retracts equivariantly onto it. By Corollary 1.1.8 we have

$$H^*_{S^1}(A) \cong H^*_{S^1}(S^1/\mu_k) \cong H^*(B\mu_k) = \mathbb{Z}[u]/(ku).$$

Now we know the middle map is $(f_N, f_S) \mapsto f_N|_A - f_S|_A$, where restricting to A means reducing modulo ku. Since $H^*_{S^1}(U_N)$, $H^*_{S^1}(U_S)$ and $H^*_{S^1}(A)$ vanish in odd degrees, we only need to consider

$$0 \to H^{2m}_{S^1}(S^2) \longrightarrow \mathbb{Z}u^m \oplus \mathbb{Z}u^m \longrightarrow \Big(\mathbb{Z}[u]/(ku)\Big)^{2m} \xrightarrow{\delta} H^{2m+1}_{S^1}(S^2) \to 0, \qquad m \geq 0.$$

Thus it is $(au^m, bu^m) \mapsto (a-b)u^m$. Notice that $\Big(\mathbb{Z}[u]/(ku)\Big)^{2m}$ is $\mathbb{Z}$ for $m = 0$ and $\mathbb{Z}/k$ for $m \geq 1$. In both cases this map is obviously surjective.

Thus $\delta = 0$, so $H_{S^1}^{\mathrm{odd}}(S^2) = 0$, and $H_{S^1}^{2m}(S^2)$ is the kernel: $\{(a,a)\} \cong \mathbb{Z}$ for $m = 0$, and $\{(a,b) : a \equiv b \bmod k\} \cong \mathbb{Z}^2$ for $m \geq 1$. Putting all degrees together,

$$H_{S^1}^*(S^2;\mathbb{Z}) \cong \{(f_N, f_S) \in \mathbb{Z}[u]^2 : f_N \equiv f_S \bmod ku\} \cong \mathbb{Z}[u,x]/\Big(x(x-ku)\Big),$$

where $x = (0, ku)$. Over $\mathbb{Q}$, ku is a unit multiple of u, so the answer is the same as for $k = 1$.

We end this section by discussing the case where G acts locally freely, that is, every stabiliser G_x is finite. Then X_G is no longer X/G, but in cohomology it still is, as long as the orders $|G_x|$ are invertible.

So far X has only been a paracompact space. For the rest of this section we assume in addition that X is a smooth G-manifold, so that the slice theorem is available. The same holds for G-CW complexes, which we do not need.

Proposition 1.1.10. *Suppose G acts locally freely on X, and let R be a ring in which every $|G_x|$ is invertible, for example $R = \mathbb{Q}$. Then the quotient map $q : X_G \to X/G$ induces an isomorphism*

$$q^* : H^*(X/G;R) \xrightarrow{\cong} H_G^*(X;R).$$

Before we prove this, we briefly recall the transfer map from [27, §3.G]. We will not explain it in detail, as it is standard algebraic topology.

Suppose $p : \widetilde{Y} \to Y$ is a d-sheeted covering. Every singular simplex $\sigma : \Delta^n \to Y$ has exactly d lifts $\widetilde{\sigma}_1, \dots, \widetilde{\sigma}_d$, that is, maps $\Delta^n \to \widetilde{Y}$ with $p \circ \widetilde{\sigma}_i = \sigma$. For a cochain φ on $\widetilde{Y}$ we set

$$(\tau\varphi)(\sigma) = \varphi(\widetilde{\sigma}_1) + \cdots + \varphi(\widetilde{\sigma}_d).$$

This commutes with the coboundary, so it gives a map $\tau : H^n(\widetilde{Y};R) \to H^n(Y;R)$.

We claim that

$$\tau \circ p^* = d \cdot \mathrm{id},$$

since for a cochain φ on Y, the cochain $p^*\varphi$ takes the value $\varphi(\sigma)$ on every lift $\widetilde{\sigma}_i$.

This gives the following lemma, which says that the fibres BG_x contribute nothing to cohomology in positive degrees.

Lemma 1.1.11. *Let H be a finite group and let R be a ring in which $|H|$ is invertible. Then $H^*(BH;R) \cong H^*(\mathrm{pt};R)$.*

Proof. Since H is finite and acts freely on EH, the map $p : EH \to BH$ is an $|H|$-sheeted covering. Let $a \in H^n(BH;R)$ with $n > 0$. Since EH is contractible, $p^*a = 0$, so

$$|H|\, a = \tau(p^*a) = 0.$$

As $|H|$ is invertible in R, we get $a = 0$. In degree 0, $H^0(BH; R) = R$ because BH is connected. □

Moreover, the slice theorem below extends this from a single fibre to a small neighbourhood of it.

Lemma 1.1.12 (Slice theorem [9, Theorem I.2.1])**.** *Let $x \in X$, and let $V_x = T_xX/T_x(G \cdot x)$, on which G_x acts linearly through the tangent maps. Then there exists a G-equivariant diffeomorphism from a G-invariant open neighbourhood of the zero section in $G \times_{G_x} V_x$ onto an open neighbourhood of $G \cdot x$ in X, which sends the zero section G/G_x onto $G \cdot x$ by the orbit map $gG_x \mapsto g \cdot x$.*

Consider the principal G_x-bundle $G \to G/G_x \cong G \cdot x \subset X$. For a small G_x-invariant disc $D \subset V_x$, Lemma 1.1.12 makes the associated disc bundle $U = G \times_{G_x} D$ a tube around $G \cdot x$ in X, see Figure 1.4.

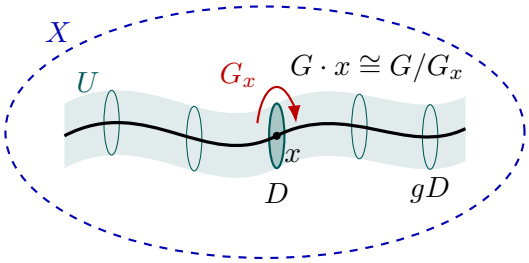


Figure 1.4: The tube U around $G \cdot x$.

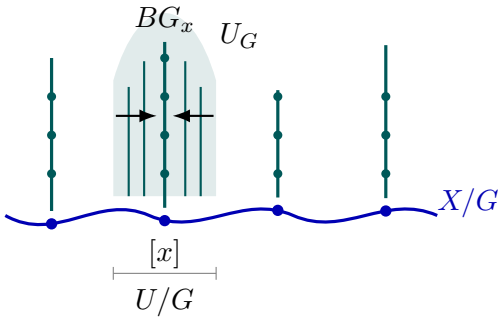


Figure 1.5: U_G retracts onto BG_x.

Scaling D to its centre retracts U equivariantly onto $G \cdot x \cong G/G_x$ and contracts $U/G = D/G_x$ to $[x]$. Hence $U_G = q^{-1}(U/G)$ retracts onto $EG \times_G G/G_x \cong BG_x$, see Figure 1.5, and by Lemma 1.1.11

$$q^* : H^*(U/G; R) \xrightarrow{\cong} H^*(U_G; R).$$

It remains to glue these local pictures together. Everything from here on is standard, and we only sketch it.

Definition 1.1.13. For an R-module A, the constant sheaf $\underline{A}_X$ on X assigns to each open set $U \subset X$ the locally constant maps $U \to A$. Let $\mathcal{F}$ be a sheaf of R-modules on X. A resolution of $\mathcal{F}$ is an exact sequence of sheaves

$$0 \to \mathcal{F} \to J^0 \to J^1 \to \cdots .$$

An injective resolution is one in which every term is injective. The sheaf cohomology of $\mathcal{F}$ is

$$H^*(X;\mathcal{F}) = H^*\Big(I^\bullet(X)\Big),$$

where $0 \to \mathcal{F} \to I^\bullet$ is an injective resolution. It does not depend on the choice of $I^\bullet$.

More generally, $H^*(X;\mathcal{F})$ can be computed from any resolution of $\mathcal{F}$ by sheaves with no higher cohomology [69, Tag 015E].

Example 1.1.14. On a manifold M, the de Rham complex

$$0 \to \underline{\mathbb{R}}_M \to \Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d} \cdots$$

is a resolution, since by the Poincaré lemma every closed form has a primitive near each point. It is not injective, but the sheaves Ω^j are fine and have no higher cohomology, so it still computes sheaf cohomology, $H^*(M;\underline{\mathbb{R}}_M) \cong H^*_{\mathrm{dR}}(M)$. After taking global sections a closed form need not have a global primitive, and this is what $H^*_{\mathrm{dR}}(M)$ measures.

In the same way the sheaves of singular cochains form a resolution of $\underline{R}_X$, and on a paracompact Hausdorff, locally contractible space X this gives a natural isomorphism $H^*(X;\underline{R}_X) \cong H^*_{\mathrm{sing}}(X;R)$, see [16, Theorem III.1.1].

This covers X/G, and also X_G for a suitable model of EG, so we may use sheaf cohomology from now on.

Proof of Proposition 1.1.10. Write $E = X_G$ and $B = X/G$, and take an injective resolution $0 \to \underline{R}_E \to I^\bullet$. Push it down to B by setting $K^j = q_* I^j$, so $K^j(W) = I^j(q^{-1}(W))$. Each K^j is injective, since q_* has the exact left adjoint q^{-1}, and

$$K^\bullet(B) = I^\bullet(E), \qquad H^j\Big(K^\bullet(W)\Big) = H^j(q^{-1}(W);R).$$

Since $G{\cdot}x$ is compact, every neighbourhood of $[x]$ contains some U/G, and $q^{-1}(U/G) = U_G$ is connected with the R-cohomology of a point. Hence on stalks $K^\bullet$ is exact in positive degrees, with $\ker(K^0 \to K^1) = \underline{R}_B$. So $K^\bullet$ is an injective resolution of $\underline{R}_B$, and

$$H^*(B;R) = H^*\Big(K^\bullet(B)\Big) = H^*\Big(I^\bullet(E)\Big) = H^*(E;R).$$

This isomorphism is q^*, since q^* comes from a map of injective resolutions of $\underline{R}_B$ into $K^\bullet$, and any such map is a homotopy equivalence. □

Remark 1.1.15. In general $q_* I^\bullet$ is not exact. Its cohomology sheaves are $R^j q_* \underline{R}_E$, and the Leray spectral sequence $E_2^{p,j} = H^p(B; R^j q_* \underline{R}_E) \Rightarrow H^{p+j}(E;R)$ assembles

$H^*(E)$ from them [16, Theorem IV.6.1]. Our case is the one where only $j = 0$ survives.

Note that the sheaf argument itself works over any ring, including $\mathbb{Z}$. The only place where we need $|G_x|$ to be invertible is Lemma 1.1.11, which makes every fibre BG_x look like a point, and once it holds everything after it runs unchanged. Over $\mathbb{Z}$ the proposition fails, for example for the annulus A in Example 1.1.9, where A/S^1 is an interval but $H^2_{S^1}(A;\mathbb{Z}) = \mathbb{Z}/k$.

1.2 Genuine and Borel equivariance

Definition 1.2.1. Let E be a generalised cohomology theory [27], for example $H^*(-;R)$ or complex K-theory. The Borel E-theory of a G-space X is

$$E^*_G(X) := E^*(X_G).$$

In other words, Borel E-theory simply applies E to the space we get from the Borel construction, and in this sense it is rather coarse.

Example 1.2.2. Let S^1 act on a point. Then $\mathrm{pt}_{S^1} = BS^1 = \mathbb{CP}^\infty$, and

$$K^0(BS^1) = \mathbb{Z}[[t-1]], \qquad K^1(BS^1) = 0,$$

where t is the class of the line bundle $ES^1 \times_{S^1} \mathbb{C}_1$, and $\mathbb{C}_m$ denotes $\mathbb{C}$ with S^1 acting by $z \mapsto \lambda^m z$. We quote this from [2, Part II]. In Section 1.4 we will compute $K(\mathbb{CP}^n)$ directly, and letting $n \to \infty$ recovers it.

The Borel construction loses information in two places. First, the Borel construction replaces the G-space X by a single space X_G.

Example 1.2.3. Let $G \neq 1$ act freely on EG and trivially on a point. Then

$$(EG)_G = EG \times_G EG \simeq BG, \qquad \mathrm{pt}_G = BG,$$

so every Borel theory treats them the same. But they are not G-homotopy equivalent. A G-homotopy equivalence $X \to Y$ restricts to a homotopy equivalence $X^H \to Y^H$ for every closed subgroup H, and here $(EG)^H = \varnothing$ while $\mathrm{pt}^H = \mathrm{pt}$ for $H \neq 1$.

A genuine equivariant theory remembers all the fixed point spaces X^H [53]. We will not need that generality.

Second, a cohomology theory may not see everything that X_G keeps. The fibres BG_x are part of X_G, but Proposition 1.1.10 says that

$$H^*_G(X;\mathbb{Q}) \cong H^*(X/G;\mathbb{Q})$$

whenever all stabilisers are finite. So rational cohomology does not see finite stabilisers at all. We saw this in Example 1.1.9. There the annulus $A \simeq S^1/\mu_k$ contributes

$$H^*_{S^1}(A;\mathbb{Z}) = H^*(B\mu_k;\mathbb{Z}) = \mathbb{Z}[u]/(ku),$$

and

$$H^*_{S^1}(S^2;\mathbb{Z}) = \{(f_N, f_S) \in \mathbb{Z}[u]^2 \mid f_N \equiv f_S \bmod ku\}.$$

Over $\mathbb{Q}$ the annulus contributes only $\mathbb{Q}$ in degree 0, just as a free orbit would, and the weight k disappears. Over $\mathbb{Z}$ it survives as torsion in A.

The whole difference between weight k and weight 1 sits in the annulus, that is, in the single orbit S^1/μ_k. So a good test for an equivariant theory is what it assigns to this orbit. By Proposition 1.1.6, a Borel theory assigns $E^*(B\mu_k)$. We now look at a theory that is not of this form, equivariant K-theory, and see whether it remembers more.

Definition 1.2.4. Let X be a compact G-space. The equivariant K-theory $K_G(X)$ is the Grothendieck group of G-equivariant complex vector bundles on X [66].

For K-theory we write $K^*(X_G)$ for the Borel theory and keep K_G for this genuine one.

Example 1.2.5. A G-bundle over a point is a representation, so $K_G(\mathrm{pt})$ is the representation ring $R(G)$. For $G = S^1$, every representation is a sum of the $\mathbb{C}_m$, and $\mathbb{C}_m \otimes \mathbb{C}_n = \mathbb{C}_{m+n}$. Hence

$$K_{S^1}(\mathrm{pt}) = R(S^1) = \mathbb{Z}[t, t^{-1}], \qquad t = [\mathbb{C}_1].$$

Compare the Borel answer $K^0(BS^1) = \mathbb{Z}[[t-1]]$ of Example 1.2.2.

Like Borel cohomology, K_G extends to a cohomology theory on compact G-spaces with Mayer–Vietoris sequences. Moreover $K_G(X) = K(X/G)$ for free actions, and $K_G(G/H) = R(H)$ for a closed subgroup H [66].

Example 1.2.6. Let S^1 act on S^2 with weight k as in Example 1.1.9. The annulus $A \simeq S^1/\mu_k$ now contributes $K_{S^1}(A) = R(\mu_k)$. The group μ_k is cyclic, so its irreducible representations are the characters $\zeta \mapsto \zeta^m$, and $\mathbb{C}_m$ restricted to μ_k depends only

on m mod k. Hence the restriction $R(S^1) \to R(\mu_k)$, $t \mapsto t$, is surjective with kernel $(t^k - 1)$, and

$$K_{S^1}(A) = R(\mu_k) = \mathbb{Z}[t]/(t^k - 1).$$

The same Mayer–Vietoris argument gives

$$K_{S^1}(S^2) = \{(f_N, f_S) \in R(S^1)^2 \mid f_N \equiv f_S \bmod t^k - 1\}.$$

Over $\mathbb{Q}$ the annulus contributes $R(\mu_k) \otimes \mathbb{Q}$, which has dimension k, so the weight k survives.

The genuine and Borel theories are related by the map

$$K_G(X) \longrightarrow K^0(X_G), \qquad V \longmapsto V_G = EG \times_G V.$$

At a point it sends $t = [\mathbb{C}_1]$ to the class t of Example 1.2.2, so it is the inclusion $\mathbb{Z}[t^{\pm 1}] \to \mathbb{Z}[[t-1]]$. Geometrically, this is the inclusion $\operatorname{Spf} \mathbb{Z}[[t-1]] \to \operatorname{Spec} \mathbb{Z}[t^{\pm 1}] = \mathbb{G}_m$ of the formal neighbourhood of 1, and in general the map is a completion.

Theorem 1.2.7 (Atiyah–Segal)**.** *Let G be a compact Lie group and X a finite G-CW complex. Then the map above induces an isomorphism from the completion of $K_G^*(X)$ at the augmentation ideal $I = \ker(R(G) \to \mathbb{Z})$ to $K^*(X_G)$ [8].*

So Borel K-theory is genuine K-theory completed at $t = 1$, and for the annulus $K^0(B\mu_k) = \mathbb{Z}[[t-1]]/(t^k - 1)$.

Over $\mathbb{C}$, taking Spec, we have $\operatorname{Spec} \mathbb{C}[u] = \mathbb{C}$ and $\operatorname{Spec} \mathbb{C}[t^{\pm 1}] = \mathbb{C}^\times$, while $\operatorname{Spf} \mathbb{C}[[t-1]]$ is the formal neighbourhood of 1. The Chern character $\mathbb{C}[[t-1]] \to \mathbb{C}[[u]]$, $t \mapsto e^u$, is an isomorphism, and on Spf it is the exponential map between the formal neighbourhoods of 0 and 1, see Figure 1.6. So the line $\operatorname{Spec} \mathbb{C}[u]$ is the Lie algebra of $\mathbb{C}^\times$.

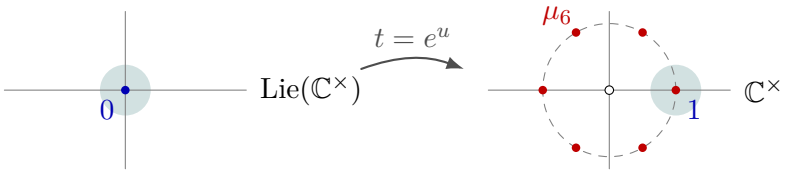


Figure 1.6: The exponential map and the points of μ_6.

For the annulus, Examples 1.1.9 and 1.2.6 give

$$H^*_{S^1}(A; \mathbb{C}) = \mathbb{C}[u]/(ku), \qquad K_{S^1}(A) \otimes \mathbb{C} = \mathbb{C}[t^{\pm 1}]/(t^k - 1).$$

The spectrum of a quotient $R/(f)$ is the zero set of f in $\operatorname{Spec} R$. These zero sets are the kernels of the weight k, seen as the linear function $d\chi_k\colon u \mapsto ku$ on the Lie algebra and as the character $\chi_k\colon t \mapsto t^k$ on the group,

$$\operatorname{Spec} H^*_{S^1}(A;\mathbb{C}) = \ker d\chi_k = \operatorname{Lie}(\mu_k) = \{0\}, \qquad \operatorname{Spec} K_{S^1}(A) \otimes \mathbb{C} = \ker \chi_k = \mu_k.$$

The subgroup μ_k consists of k isolated points, so its tangent space at the identity is zero. So the annulus contributes one dimension in rational cohomology, just like a free orbit, but k dimensions in genuine K-theory. This is why the weight k is lost in Example 1.1.9.

To sum up, in both theories the annulus is cut out by a single relation, ku or $t^k - 1$. Its zero set is $\operatorname{Lie}(\mu_k)$ in rational cohomology and μ_k in genuine K-theory. In the next section we will see that this relation is, up to a unit, the Euler class of the line $\mathbb{C}_k$.

1.3 Borel–Moore homology and Euler classes

The cap U_N of Example 1.1.9 is equivariantly diffeomorphic to the tangent line $\mathbb{C}_k$ at N, and the annulus A is equivariantly homotopy equivalent to U_N minus the pole, that is, to $\mathbb{C}_k \setminus 0 \simeq S^1/\mu_k$. So the question left by Section 1.2 is what happens to equivariant cohomology and K-theory when we remove the origin from $\mathbb{C}_k$, or more generally a closed submanifold from a manifold.

We will see that we pay the Euler class of its normal bundle. To say this we first need classes of closed submanifolds that are not compact.

Definition 1.3.1. Let X be the interior of a compact manifold W with boundary, so that its one-point compactification is $\bar{X} = X \cup \{\infty\} = W/\partial W$. The Borel–Moore homology of X is $H^{BM}_*(X) = H_*(\bar{X}, \infty) = H_*(W, \partial W)$ [20, §1].

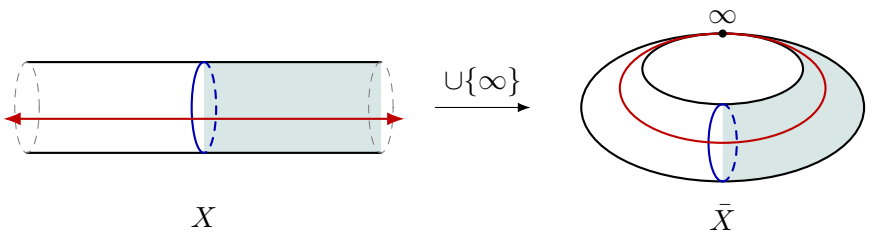


Figure 1.7: In $\bar{X}$ the line becomes a circle and the core circle bounds a disc.

Definition 1.3.1 is short, but it does not say what a Borel–Moore cycle is. At the chain level, $H^{BM}_*(X)$ is equivalently the homology of locally finite chains, that is,

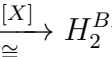
possibly infinite sums of simplices that meet each compact set only finitely often [20, §1]. So a Borel–Moore cycle may run off to infinity. An oriented submanifold $Z \subset X$ that is closed as a subset has a fundamental class $[Z] \in H^{BM}_{\dim Z}(X)$, whether it is compact or not. In particular X itself has a fundamental class $[X]$, and Poincaré duality below is cap product with it.

Theorem 1.3.2 (Poincaré duality)**.** *If moreover W is oriented of dimension n, there is a canonical isomorphism $H^k(X) \cong H^{BM}_{n-k}(X)$ [20, §1], [27, Theorem 3.43].*

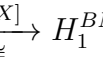
Remark 1.3.3. For oriented W, Poincaré duality can itself be taken as the definition of $H^{BM}_*(X)$. In fact each of the four descriptions above can serve as an equivalent definition [20, §1]. The space $\bar{X} = W/\partial W$ gives a geometric picture, see Figure 1.7.

Since $H^k(X) = H^k(W)$, Theorem 1.3.2 is Lefschetz duality for the manifold with boundary W, namely $H^k(W) \cong H_{n-k}(W, \partial W)$.

Example 1.3.4. Let $X = \mathbb{C}^\times$, an open cylinder  . Poincaré duality sends the constant function to the fundamental class,

$$H^0(\text{  }) \xrightarrow[\cong]{\cap[X]} H_2^{BM}(\text{  }), \qquad 1 \longmapsto [\text{  }].$$

In degree one it sends the generator $d\theta/2\pi$ of $H^1(X) = \mathbb{Z}$ to the line $\mathbb{R}_{>0}$,

$$H^1(\text{  }) \xrightarrow[\cong]{\cap[X]} H_1^{BM}(\text{  }), \qquad \frac{d\theta}{2\pi} \longmapsto [\text{  }].$$

To see this, write $X = S^1 \times \mathbb{R}$ by $z = e^{t+i\theta}$, so that $[\text{  }] = [S^1] \times [\mathbb{R}]$. The form $d\theta/2\pi$ is pulled back from the circle factor, so capping with it integrates over S^1 and leaves the line $\{\theta = 0\} \times \mathbb{R} = \mathbb{R}_{>0}$,

$$\frac{d\theta}{2\pi} \cap [\text{  }] = \frac{d\theta}{2\pi} \cap \big([S^1] \times [\mathbb{R}]\big) = \Big(\int_{S^1} \frac{d\theta}{2\pi}\Big)[\text{  }] = [\text{  }].$$

The core circle generates $H_1(X)$, but it is zero in $H_1^{BM}(X)$, since it bounds the half cylinder $\{|z| \geq 1\}$, which runs off to infinity, see Figure 1.7,

$$\text{  } = \partial \text{  }, \qquad \text{so} \qquad [\text{  }] = 0 \in H_1^{BM}(\text{  }).$$

We now pass to the equivariant setting, in the Borel sense. From now on T is a torus acting smoothly on a manifold M. The space M_T is usually infinite-dimensional, so the Poincaré duality of Theorem 1.3.2 has no ambient dimension to work with. We therefore replace ET by a finite-dimensional approximation, a free T-manifold E whose reduced cohomology vanishes in a large range of degrees, with $B = E/T$ an

oriented manifold. For $T = S^1$ we may take $E = S^{2m+1}$ with the scalar action, and $B = \mathbb{CP}^m$. For E large compared with the degree, we define

$$H_i^{BM,T}(M) := H^{BM}_{i+\dim_{\mathbb{R}} B}(E \times_T M)$$

[3, §17.1]. The shift accounts for the base of the bundle $M \to E \times_T M \to B$.

If M is an oriented T-manifold of real dimension n, then $E \times_T M$ is an oriented manifold of dimension $n + \dim_{\mathbb{R}} B$, and Poincaré duality gives, in the stable range,

$$H_T^k(M) \cong H^k(E \times_T M) \cong H^{BM}_{n+\dim_{\mathbb{R}} B-k}(E \times_T M) = H^{BM,T}_{n-k}(M).$$

In particular, let $Z \subset M$ be a T-invariant oriented submanifold of dimension d, closed as a subset. Then $E \times_T Z$ is a closed oriented submanifold of $E \times_T M$ (oriented since T is connected), which gives a class

$$[Z]_T := [E \times_T Z] \in H_d^{BM,T}(M) \cong H_T^{n-d}(M).$$

It does not depend on the choice of E, and $[M]_T = 1$.

We include equivariant Borel–Moore homology for completeness, since it gives Poincaré duality in the equivariant setting. The Thom class, which is closely tied to Poincaré duality, needs much less. If $V \to Z$ is a T-equivariant complex vector bundle of rank r, then $V_T \to Z_T$ is a complex vector bundle of rank r, and every characteristic class of V_T is an equivariant characteristic class of V, living in $H^*(Z_T) = H_T^*(Z)$.

Definition 1.3.5. The Thom class $\mathrm{Th}(V) \in H_T^{2r}(V, V \setminus Z)$ is the Thom class of V_T, the class that restricts on each fibre to the generator of $H^{2r}(\mathbb{C}^r, \mathbb{C}^r \setminus 0)$. The Euler class $e_T(V) \in H_T^{2r}(Z)$ is its restriction to the zero section.

By the Thom isomorphism, cup product with $\mathrm{Th}(V)$ gives $H_T^*(Z) \cong H_T^{*+2r}(V, V \setminus Z)$, and $e_T(V)$ is the usual Euler class of V_T [54, Ch. 23, §§5, 7]. The Thom and Euler classes have the following geometric meaning. We only sketch the argument, and only over $\mathbb{R}$.

Proposition 1.3.6. *Suppose Z is oriented, so that the total space V is oriented too. (a) The image of $\mathrm{Th}(V)$ in $H_T^{2r}(V)$ is the class $[Z]_T$ of the zero section. (b) If s is a T-equivariant section of V that meets the zero section transversally, then $e_T(V) = [s^{-1}(0)]_T$ in $H_T^{2r}(Z)$, with the orientation of $s^{-1}(0)$ induced by s.*

Proof idea. On the approximations $E \times_T V \to E \times_T Z$ both statements are non-equivariant, so we forget the action. We use real coefficients and differential forms. The class of a closed oriented submanifold $Y \subset M$ of dimension d is characterised by

$$\int_Y \alpha = \int_M \alpha \wedge [Y]$$

for every closed compactly supported d-form α on M [14]. The Thom class is represented by a closed form supported near the zero section whose integral over each fibre is 1. Near the zero section we may replace α by the pull-back of its restriction to Z, and integrating over the fibres first leaves $\int_Z \alpha$. This is (a).

For (b), the sections λs with $0 \leq \lambda \leq 1$ join s to the zero section, so $e(V) = s^*\mathrm{Th}(V)$. This form is supported near $s^{-1}(0)$. Since s is transverse, it maps each small normal slice to $s^{-1}(0)$ diffeomorphically onto a neighbourhood of 0 in the fibre, so $s^*\mathrm{Th}(V)$ has integral 1 over each normal slice, and the argument for (a) shows that it is $[s^{-1}(0)]$. So only the zeros of s are counted, see Figure 1.8. □

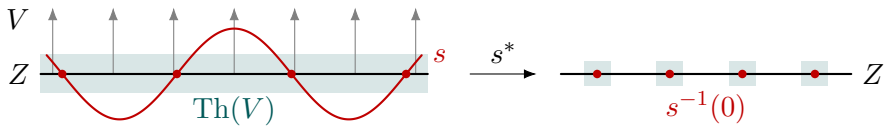


Figure 1.8: The Euler class is the zero set of a generic section.

Whenever Euler classes appear, the Gysin map appears too. Let $Z \subset M$ be an invariant submanifold, closed as a subset, whose normal bundle ν is a T-equivariant complex vector bundle of rank c, and let $i\colon Z \to M$ be the inclusion. Unlike the pullback i^*, the Gysin map goes in the direction of the embedding. Choose an invariant tubular neighbourhood of Z, identify it with ν, and let $\pi\colon \nu \to Z$ be the projection.

Definition 1.3.7. The *Gysin map* $i_!\colon H_T^*(Z) \to H_T^{*+2c}(M)$ is the composite

$$H_T^*(Z) \xrightarrow{\pi^*} H_T^*(\nu) \xrightarrow{\smile \mathrm{Th}(\nu)} H_T^{*+2c}(\nu, \nu \setminus Z) \cong H_T^{*+2c}(M, M \setminus Z) \longrightarrow H_T^{*+2c}(M),$$

where the isomorphism is excision and the last map forgets the support condition.

So $i_!$ takes a class on Z, spreads it in the normal direction by the Thom class, and places it inside M, see Figure 1.9. With differential forms, $H_T^*(\nu, \nu \setminus Z)$ is represented by forms with compact support in the normal direction, and the last map is extension by zero.

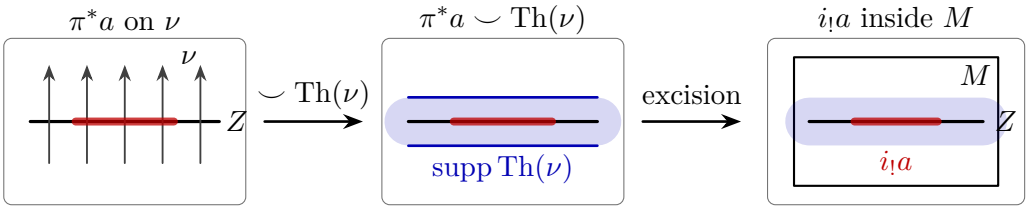


Figure 1.9: The Gysin map $i_!$.

Corollary 1.3.8 (Self-intersection formula)**.** *For $a \in H_T^*(Z)$ we have $i^* i_! a = a \cdot e_T(\nu)$. If M is oriented and Z is oriented compatibly with M and ν, then $i_! 1 = [Z]_T$.*

Proof. The restriction i^* only sees the tubular neighbourhood, where i is the zero section s_0 of ν. Since $\pi \circ s_0 = \mathrm{id}$,

$$i^* i_! a = s_0^*\Big(\pi^* a \smile \mathrm{Th}(\nu)\Big) = a \smile s_0^* \mathrm{Th}(\nu) = a \cdot e_T(\nu).$$

The second statement is Proposition 1.3.6(a), moved into M by excision. □

Corollary 1.3.9. *If V has a nowhere vanishing T-equivariant section, then $e_T(V) = 0$.*

Proof. A nowhere vanishing section s has $s(Z) \subset V \setminus Z$, so

$$\begin{array}{ccc} H_T^{2r}(V, V \setminus Z) & \longrightarrow & H_T^{2r}(V) \\ {\scriptstyle s^*}\downarrow & & \downarrow{\scriptstyle s^*} \\ H_T^{2r}(Z, Z) = 0 & \longrightarrow & H_T^{2r}(Z) \end{array}$$

sends $\mathrm{Th}(V)$ to $s^*\mathrm{Th}(V) = e_T(V)$ through 0. □

Without the action, the Euler class is the obstruction to a nowhere vanishing section. For $\mathbb{C}_k$ over a point, any nonzero vector v is such a section, so $e(\mathbb{C}_k) = 0$. An equivariant section must satisfy $\lambda \cdot v = v$, that is $\lambda^k v = v$, for every $\lambda \in S^1$, and for $k \neq 0$ this forces $v = 0$. So the obstruction can survive equivariantly, and indeed $e_{S^1}(\mathbb{C}_k) = ku \neq 0$ (Example 1.3.10).

Example 1.3.10. Let S^1 act on $\mathbb{C}_k$, a vector bundle over a point. Its total space is the plane $\mathbb{C}_k$, its zero section is the origin $i\colon 0 \to \mathbb{C}_k$, and the normal bundle of the origin is $\mathbb{C}_k$ itself. So Corollary 1.3.8 gives $e_{S^1}(\mathbb{C}_k) = i^*[0]_{S^1}$, and since $\mathbb{C}_k$ contracts to the origin, i^* is an isomorphism. So the Euler class is just the class of the origin,

$$e_{S^1}(\mathbb{C}_k) = [0]_{S^1}.$$

From the Borel–Moore point of view, without the action $[0] \in H_0^{BM}(\mathbb{C})$ is minus the boundary of the ray $R = \mathbb{R}_{\geq 0}$, which runs off to infinity, so $e(\mathbb{C}_k) = [0] = 0$, see Figure 1.10. With the action, $[0]_{S^1}$ is the class of the zero section BS^1 of $L_k = ES^1 \times_{S^1} \mathbb{C}_k$. The natural way to kill it is a ray in every fibre, chosen continuously, that is, a nowhere vanishing section of L_k. Over $\mathbb{CP}^1 \subset BS^1$ the rays chosen on two discs differ on the equator by $\lambda \mapsto \lambda^k$, so one turns k times relative to the other. This k is $c_1(L_k)|_{\mathbb{CP}^1}$, which we now compute.

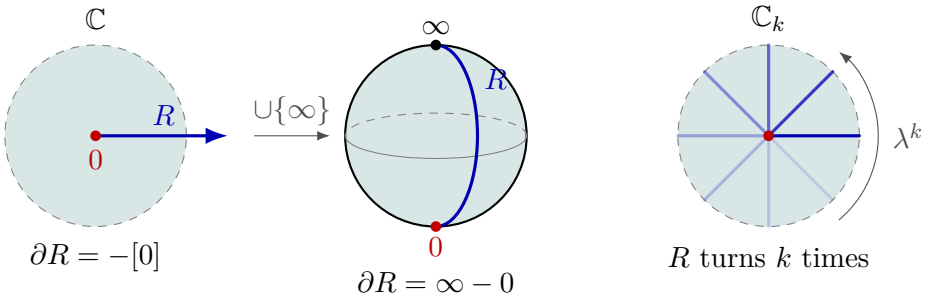


Figure 1.10: The origin bounds the ray R, and with the action R turns k times.

To compute, write $L_m = ES^1 \times_{S^1} \mathbb{C}_m$ and set $u = c_1(L_1) \in H^2(BS^1)$. Then $(\mathbb{C}_k)_{S^1} = L_1^{\otimes k}$, so $e_{S^1}(\mathbb{C}_k) = c_1(L_1^{\otimes k}) = ku$. For weights $w_1, \dots, w_n$, the Euler class of a sum is the product, so

$$e_{S^1}\Big(\bigoplus_j \mathbb{C}_{w_j}\Big) = \Big(\prod_j w_j\Big)u^n. \tag{1.3.1}$$

A coordinate subspace $\mathbb{C}^I$ has normal bundle $\mathbb{C}^I \times \bigoplus_{j\notin I} \mathbb{C}_{w_j}$, and restriction from $\mathbb{C}^n$ to $\mathbb{C}^I$ is an isomorphism in $H^*_{S^1}$, as both contract to the origin. So Corollary 1.3.8 and (1.3.1) give

$$[\mathbb{C}^I]_{S^1} = \Big(\prod_{j\notin I} w_j\Big)u^{n-|I|},$$

and in particular $[0]_{S^1} = \big(\prod_j w_j\big)u^n$.

The same picture holds in algebraic K-theory, with coherent sheaves in place of cycles. We first introduce equivariant coherent sheaves.

Definition 1.3.11. Let T be a compact torus acting linearly on a complex vector space V, and equip $A = \mathbb{C}[V]$ with the action $(\lambda \cdot f)(v) = f(\lambda^{-1} \cdot v)$.

Under the affine sheaf–module correspondence, a *T-equivariant coherent sheaf* on V is a finitely generated A-module M with a locally finite T-action satisfying $\lambda \cdot (fm) = (\lambda \cdot f)(\lambda \cdot m)$.

Here locally finite means that every element lies in a finite-dimensional continuous T-subrepresentation. For $T = S^1$ this says $M = \bigoplus_{m\in\mathbb{Z}} M_m$, with λ acting on M_m by λ^m, just as A splits into monomials. The compatibility then says that a function of weight a maps M_b to M_{a+b}. So M is simply a graded A-module, and for a general torus the grading is by the character lattice.

We use finite equivariant vector-bundle resolutions to represent coherent sheaves in K-theory. On the representation space V, algebraic and topological equivariant K-theory both identify with $R(T)$, and under this identification $[\mathcal{O}_V \otimes W] = [W]$.

Example 1.3.12. Let $T = S^1$, write $t = [\mathbb{C}_1]$, and take $V = \bigoplus_{j=1}^n \mathbb{C}_{w_j}$. The coordinate z_j has weight $-w_j$, and the origin $i\colon \{0\} \hookrightarrow V$ determines the skyscraper sheaf

$$\mathcal{O}_0 \longleftrightarrow A/(z_1, \dots, z_n) \cong \mathbb{C},$$

with trivial S^1-action. For one coordinate, the sequence

$$0 \longrightarrow \mathcal{O}_V \otimes \mathbb{C}_{-w} \xrightarrow{z} \mathcal{O}_V \longrightarrow \mathcal{O}_0 \longrightarrow 0$$

is exact and equivariant, giving $[\mathcal{O}_0] = 1 - t^{-w}$.

In general, $z_1, \dots, z_n$ form a regular sequence, so the tensor product of the corresponding two-term complexes gives the Koszul resolution

$$0 \longrightarrow \mathcal{O}_V \otimes \Lambda^n V^\vee \longrightarrow \cdots \longrightarrow \mathcal{O}_V \otimes V^\vee \longrightarrow \mathcal{O}_V \longrightarrow \mathcal{O}_0 \longrightarrow 0.$$

Taking its alternating sum yields

$$i_!1 = [\mathcal{O}_0] = \sum_{q=0}^{n} (-1)^q [\Lambda^q V^\vee] = \prod_{j=1}^{n} (1 - t^{-w_j}).$$

Since V equivariantly contracts to the origin, restriction $i^*\colon K_{S^1}(V) \to R(S^1)$ is an isomorphism. We define the K-theory Euler class by the self-intersection formula, $e_K(V) := i^* i_! 1$. Pulling back the vector-bundle terms gives $e_K(V) = \sum_q (-1)^q [\Lambda^q V^\vee]$.

With u as in Example 1.3.10, the Chern character takes values in $\mathbb{Q}[[u]]$ and sends t to e^u. Hence

$$\operatorname{ch}(e_K(V)) = \prod_{j=1}^{n} (1 - e^{-w_j u}) = \Big(\prod_{j=1}^{n} w_j \Big) u^n + O(u^{n+1}).$$

Its degree-$2n$ component is the cohomological Euler class in (1.3.1).

Return to Definition 1.3.7 and put $U = M \setminus Z$. Excision and the Thom isomorphism give $H_T^*(M, U) \cong H_T^*(\nu, \nu \setminus Z) \cong H_T^{*-2c}(Z)$, so the long exact sequence of the pair (M, U) becomes the Gysin sequence

$$\cdots \to H_T^{*-2c}(Z) \xrightarrow{i_!} H_T^*(M) \longrightarrow H_T^*(U) \longrightarrow H_T^{*-2c+1}(Z) \to \cdots,$$

whose first map is the pushforward $i_!$. By Corollary 1.3.8, $i^* i_!$ is cup product with $e_T(\nu)$, so whether this sequence breaks into short exact sequences is decided by the Euler class.

Lemma 1.3.13. *If $e_T(\nu)$ is not a zero divisor in $H_T^*(Z)$, then*

$$0 \to H_T^{*-2c}(Z) \xrightarrow{i_!} H_T^*(M) \to H_T^*(U) \to 0$$

is exact, so $H_T^*(U) = H_T^*(M)/i_! H_T^*(Z)$.

Proof. By Corollary 1.3.8, $i^*i_!$ is multiplication by $e_T(\nu)$, which is injective, so $i_!$ is injective in every degree. Hence the connecting map $H_T^*(U) \to H_T^{*-2c+1}(Z)$ is zero, and the Gysin sequence breaks into these short exact sequences. □

Example 1.3.14. Take $M = \mathbb{C}_k$ and $Z = 0$. Since $\mathbb{C}_k$ contracts equivariantly to the origin, i^* is an isomorphism, so $i_!a$ corresponds to $a \cdot ku$ and the image of $i_!$ is the ideal $(ku) \subset \mathbb{Z}[u]$. The class ku is not a zero divisor, and $U = \mathbb{C}_k \setminus 0 \simeq S^1/\mu_k$, so

$$H^*_{S^1}(S^1/\mu_k) = \mathbb{Z}[u]/(ku).$$

The same argument works in genuine K-theory, with the Euler class $e_K(\mathbb{C}_k) = 1 - t^{-k}$ of Example 1.3.12 and Segal's Thom isomorphism [66, Proposition 3.2], and gives

$$K_{S^1}(S^1/\mu_k) = R(S^1)/(1 - t^{-k}) = R(\mu_k).$$

This is the annulus of Examples 1.1.9 and 1.2.6, and its relation is, up to the unit t^k in K-theory, the Euler class of the tangent line at the north pole.

Example 1.1.9 can also be seen in this language. The tangent weights at N and S are k and $-k$, so by Corollary 1.3.8 the classes of the two poles restrict to $(ku, 0)$ and $(0, -ku)$, and $[N]_{S^1} - [S]_{S^1} = ku \cdot 1$. The two poles are homologous, but equivariantly they differ by ku.

The Thom isomorphism, the Gysin sequence, the self-intersection formula and Lemma 1.3.13 only use Thom classes of complex vector bundles. So they hold in any multiplicative cohomology theory E in which complex vector bundles have Thom classes. To see what this asks for, start with line bundles. By Proposition 1.1.3, a line bundle $L \to X$ is a map $f_L \colon X \to \mathbb{CP}^\infty$, so a natural first Chern class in E is one class $u \in \widetilde{E}^2(\mathbb{CP}^\infty)$, with $c_1^E(L) = f_L^* u$. In ordinary cohomology c_1 of the tautological bundle generates $H^2(\mathbb{CP}^1) = H^2(S^2)$, and we ask the same in E.

Definition 1.3.15. A *complex orientation* of a multiplicative cohomology theory E is a class $u \in \widetilde{E}^2(\mathbb{CP}^\infty)$ whose restriction to $\mathbb{CP}^1 = S^2$ is the unit $1 \in \widetilde{E}^2(S^2) \cong E^0(\mathrm{pt})$. We then call E *complex oriented*, and set $c_1^E(L) = f_L^* u$.

The Thom space of the tautological bundle on $\mathbb{CP}^\infty$ is homotopy equivalent to $\mathbb{CP}^\infty$, and a fibre has Thom space $\mathbb{CP}^1$, so u is exactly a Thom class of the tautological bundle. It then extends to all complex vector bundles.

Theorem 1.3.16 ([2, Part II])**.** *Let E be complex oriented and homotopy commutative. Every complex vector bundle $V \to Z$ of rank r has a Thom class $\mathrm{Th}^E(V) \in E^{2r}(V, V \setminus Z)$, restricting to the unit on each fibre, natural under pull-back, and with $\mathrm{Th}^E(V \oplus W) = \mathrm{Th}^E(V)\,\mathrm{Th}^E(W)$. Its restriction $e^E(V)$ to the zero section is the* Euler class*, and $e^E(L) = c_1^E(L)$ for a line bundle.*

Applied to the bundles V_T, these Thom classes make the Borel theory E_T of Definition 1.2.1 satisfy the Thom isomorphism, the Gysin sequence, the self-intersection formula and Lemma 1.3.13, with the same proofs. Examples are ordinary cohomology $H^*(-;R)$, complex K-theory and the Morava K-theories of Chapter 3.

1.4 Complex projective space

Lemma 1.3.13 answers the question raised at the start of Section 1.3. When we remove a closed invariant submanifold, the equivariant cohomology pays the Euler class of its normal bundle.

We now compute a familiar example, complex projective space. Let S^1 act on $\mathbb{C}^{n+1}$ by scalars. The action on $\mathbb{C}^{n+1}\setminus 0$ is free, and $z\mapsto([z],|z|)$ identifies the quotient with $\mathbb{CP}^n\times(0,\infty)$, so by Corollary 1.1.7

$$H^*_{S^1}(\mathbb{C}^{n+1}\setminus 0)\cong H^*(\mathbb{CP}^n).$$

Example 1.4.1. Here $M=\mathbb{C}^{n+1}$ contracts to $Z=0$, so $H^*_{S^1}(\mathbb{C}^{n+1})=\mathbb{Z}[u]$. The normal bundle of the origin is $\mathbb{C}^{n+1}$ with all weights one, so its Euler class is u^{n+1} by (1.3.1), which is not a zero divisor in $\mathbb{Z}[u]$. As in Example 1.3.14, the image of $i_!$ is (u^{n+1}), and restriction is a ring map, so Lemma 1.3.13 gives the ring isomorphism

$$H^*(\mathbb{CP}^n)\cong\mathbb{Z}[u]/(u^{n+1}).$$

The restriction $H^*_{S^1}(\mathbb{C}^{n+1})\to H^*(\mathbb{CP}^n)$ is the first example of a Kirwan map [44]. It is onto because the Euler class we pay is not a zero divisor, and Chapter 2 runs the same argument stratum by stratum.

Example 1.4.2. The same computation works in genuine K-theory, with the unit sphere S^{2n+1} in place of $\mathbb{C}^{n+1}\setminus 0$ as in Example 1.3.14. Now $K_{S^1}(\mathbb{C}^{n+1})=R(S^1)=\mathbb{Z}[t^{\pm1}]$, and by Example 1.3.12 the origin has Euler class $(1-t^{-1})^{n+1}=t^{-n-1}(t-1)^{n+1}$, which is not a zero divisor since $R(S^1)$ is a domain. The action on S^{2n+1} is free, so $K_{S^1}(S^{2n+1})=K(\mathbb{CP}^n)$, and as t is a unit modulo $(t-1)^{n+1}$ we get

$$K^0(\mathbb{CP}^n)\cong\mathbb{Z}[t]/\big((t-1)^{n+1}\big),\qquad K^1(\mathbb{CP}^n)=0,$$

with t the class of $S^{2n+1}\times_{S^1}\mathbb{C}_1$. Here K^1 vanishes because $K^1_{S^1}(\mathrm{pt})=0$. Under $t\mapsto e^u$ the relation $(t-1)^{n+1}$ becomes a unit times u^{n+1}, as in Example 1.4.1.

Letting $n\to\infty$, the restriction maps are onto and every $K^1(\mathbb{CP}^n)$ vanishes, so the $\varprojlim^1$ terms of the Milnor sequence vanish and

$$K^0(BS^1)=\varprojlim_n\mathbb{Z}[t]/\big((t-1)^{n+1}\big)=\mathbb{Z}[[t-1]],\qquad K^1(BS^1)=0,$$

as quoted in Example 1.2.2.

Now let a torus T act on $\mathbb{C}^{n+1}$ through characters $\chi_0, \dots, \chi_n$, that is $g \cdot z = (\chi_0(g)z_0, \dots, \chi_n(g)z_n)$. This commutes with scalars, so it descends to $\mathbb{CP}^n$, $g \cdot [z_0 : \cdots : z_n] = [\chi_0(g)z_0 : \cdots : \chi_n(g)z_n]$.

Proposition 1.4.3. *We write the group of characters additively, so $\chi_j - \chi_i$ is the character $g \mapsto \chi_j(g)\chi_i(g)^{-1}$. Put $a_i = e_T(\mathbb{C}_{\chi_i}) \in H^2(BT;\mathbb{Z})$, let $x = c_1^T(\mathcal{O}(1))$, where $\mathcal{O}(1)$ is the dual of the tautological line bundle $\mathcal{O}(-1) \subset \mathbb{CP}^n \times \mathbb{C}^{n+1}$ with the action induced from $\mathbb{C}^{n+1}$, and let $P_i = [e_i]$ be the coordinate points.*

(a) $H_T^*(\mathbb{CP}^n;\mathbb{Z}) = H^*(BT;\mathbb{Z})[x] \big/ \prod_{i=0}^n (x + a_i)$. It is a free $H^*(BT;\mathbb{Z})$-module with basis $1, x, \dots, x^n$, and also with the Newton basis $N_0 = 1$, $N_m = \prod_{l<m}(x + a_{\sigma(l)})$ for $1 \le m \le n$, for any ordering σ of $\{0, \dots, n\}$.

(b) Each P_i is fixed, $x|_{P_i} = -a_i$, and $T_{P_i}\mathbb{CP}^n = \bigoplus_{j \ne i} \mathbb{C}_{\chi_j - \chi_i}$. The class N_m vanishes at $P_{\sigma(0)}, \dots, P_{\sigma(m-1)}$ and equals $\prod_{l<m}(a_{\sigma(l)} - a_{\sigma(m)})$ at $P_{\sigma(m)}$.

(c) If the χ_i are pairwise distinct, the fixed points are exactly $P_0, \dots, P_n$.

(d) For a circle acting with weights $k_0, \dots, k_n$, $H_{S^1}^*(\mathbb{CP}^n;\mathbb{Z}) = \mathbb{Z}[u,x] / \prod_i (x + k_i u)$ and $x|_{P_i} = -k_i u$.

Proof. (a) Let $T' = S^1 \times T$ act on $\mathbb{C}^{n+1}$ by $(\lambda, g) \cdot z = \lambda(g \cdot z)$, and put $X = \mathbb{C}^{n+1} \setminus 0$. With $ET' = ES^1 \times ET$ we have $X_{T'} = ET \times_T (ES^1 \times_{S^1} X)$, and since S^1 acts freely, $ES^1 \times_{S^1} X \to X/S^1 \simeq \mathbb{CP}^n$ is a T-map with contractible fibres. Hence $H_{T'}^*(X) \cong H_T^*(\mathbb{CP}^n)$.

Now $H_{T'}^*(\mathbb{C}^{n+1}) = H^*(BT)[u_0]$, with u_0 the class u of the factor S^1. The normal space of the origin is $\bigoplus_i \mathbb{C}_{(1,\chi_i)}$, and first Chern classes add under tensor product, so its Euler class is $\prod_i (u_0 + a_i)$. This is monic in u_0, hence not a zero divisor, and Lemma 1.3.13 gives $H^*(BT)[u_0] / \prod_i (u_0 + a_i)$, as in Example 1.4.1.

The class u_0 is c_1 of the Borel construction of $X \times \mathbb{C}_{(1,0)}$, which descends along $X \to X/S^1$ to $(X \times \mathbb{C})/S^1$, with $\lambda(z,v) = (\lambda z, \lambda v)$ and $g[z,v] = [gz, v]$. Sending $[z,v]$ to the linear form on $\mathbb{C}z$ with $z \mapsto v$ identifies it equivariantly with $\mathcal{O}(1)$, so $u_0 \mapsto x$. The relation is monic of degree $n+1$ in x, so $1, x, \dots, x^n$ is a basis, and the Newton basis differs from it by a unitriangular matrix.

(b) We have $g \cdot [e_i] = [\chi_i(g)e_i] = [e_i]$. The fibre of $\mathcal{O}(1)$ at P_i is $(\mathbb{C}e_i)^\vee$, of weight $-\chi_i$, so $x|_{P_i} = -a_i$. The tangent space $\operatorname{Hom}(\mathbb{C}e_i, \mathbb{C}^{n+1}/\mathbb{C}e_i)$ has weights $\chi_j - \chi_i$, $j \ne i$. Restriction to P_i is a ring map, so $N_m|_{P_i} = \prod_{l<m}(a_{\sigma(l)} - a_i)$, which gives the claims about N_m.

(c) The point $[z]$ is fixed exactly when $\mathbb{C}z$ is invariant, that is, when T acts on it by some character χ. Then $z_j \neq 0$ only when $\chi_j = \chi$, so for pairwise distinct χ_j only one coordinate is non-zero.

(d) Here $a_i = k_i u$ by Example 1.3.10. □

We now look at a variant of complex projective space, which is an orbifold.

Definition 1.4.4. Let $k_0, \dots, k_n \geq 1$, and let S^1 act on $\mathbb{C}^{n+1}$ with these weights, $\lambda \cdot z = (\lambda^{k_0} z_0, \dots, \lambda^{k_n} z_n)$. The *weighted projective space* is

$$\mathbb{CP}(k_0, \dots, k_n) = S^{2n+1}/S^1.$$

The stabiliser of z is μ_d with $d = \gcd\{k_i : z_i \neq 0\}$, so the action is locally free, and it is not free as soon as some $k_i > 1$. For example, e_n has stabiliser μ_{k_n}. By Lemma 1.1.12, $\mathbb{CP}(k_0, \dots, k_n)$ is therefore an orbifold, locally $\mathbb{C}^n/\mu_d$, and $H^*_{S^1}(S^{2n+1})$ is the cohomology of this orbifold, which remembers the stabilisers. By Proposition 1.1.10, $H^*_{S^1}(S^{2n+1}; R) \cong H^*(\mathbb{CP}(k_0, \dots, k_n); R)$ when every k_i is invertible in R. Over $\mathbb{Z}$ the two differ, as we now compute.

Example 1.4.5. The proof of Example 1.4.1 applies, since S^{2n+1} is an equivariant deformation retract of $\mathbb{C}^{n+1} \setminus 0$. Now the origin has Euler class $(k_0 \cdots k_n)u^{n+1}$ by (1.3.1), which is still not a zero divisor in $\mathbb{Z}[u]$, so

$$H^*_{S^1}(S^{2n+1}; \mathbb{Z}) = \mathbb{Z}[u] \Big/ \big((k_0 \cdots k_n)u^{n+1}\big).$$

This has $\mathbb{Z}/(k_0 \cdots k_n)$ in every even degree above $2n$, while the space $\mathbb{CP}(k_0, \dots, k_n)$, of dimension $2n$, has no cohomology there. Over $\mathbb{Q}$ the torsion disappears and we get $\mathbb{Q}[u]/(u^{n+1})$, as for $\mathbb{CP}^n$.

In genuine K-theory, the proof of Example 1.4.2 gives $K^1_{S^1}(S^{2n+1}) = 0$ and

$$K^0_{S^1}(S^{2n+1}) = R(S^1) \Big/ \prod_i (1 - t^{-k_i}) = \mathbb{Z}[t] \Big/ \prod_i (t^{k_i} - 1),$$

free of rank $k_0 + \cdots + k_n$, since $\prod_i (t^{k_i} - 1)$ is monic with constant term ± 1. Its complex zeros are the elements of the stabilisers μ_{k_i}, so genuine K-theory sees every element of every stabiliser, while rational cohomology sees only the identity.

Example 1.4.6. Let p be a prime. The *teardrop* is $\mathbb{CP}(1, p)$. The point $[0 : 1]$ has stabiliser μ_p, and all other points have trivial stabiliser. The coarse space is a sphere, since $[z_0 : z_1] \mapsto [z_0^p : z_1]$ is a continuous bijection from the compact space $\mathbb{CP}(1, p)$ onto $\mathbb{CP}^1$. By Example 1.4.5, $H^*_{S^1}(S^3; \mathbb{Z}) = \mathbb{Z}[u]/(pu^2)$, of rank 2 over $\mathbb{Q}$,

and $K^0_{S^1}(S^3)$ is free of rank $1+p$, see Figure 1.11. Over $\mathbb{F}_p$ the Euler class pu^2 vanishes, so $i_! = 0$, and the Gysin sequence gives $\mathbb{F}_p$ in degrees $0, 2, 3, 4, \ldots$. As the coarse space is a sphere, Proposition 1.1.10 fails over $\mathbb{F}_p$, so Chapter 3 keeps the Borel target of Definition 2.1.4.

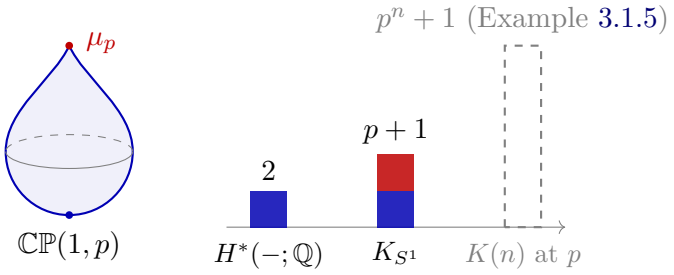


Figure 1.11: The teardrop in three theories, drawn for $p = 3$ and height $n = 2$.

Remark 1.4.7. Consider the weighted projective lines $\mathbb{CP}(1,6)$ and $\mathbb{CP}(2,3)$. The first has one point with stabiliser μ_6, and the second has points with stabilisers μ_2 and μ_3. By Example 1.4.5 both have the same integral Borel cohomology $\mathbb{Z}[u]/(6u^2)$, but their genuine K-theories have ranks $1+6=7$ and $2+3=5$. The two extra zeros of $(t-1)(t^6-1)$ are the primitive sixth roots of unity, the elements of order 6, which lie in μ_6 but in neither μ_2 nor μ_3. Borel K-theory, the completion at $t=1$ (Theorem 1.2.7), does not see them either. In $\mathbb{Z}[[t-1]]$ the two relations generate the same ideal, since

$$\frac{(t-1)(t^6-1)}{(t^2-1)(t^3-1)} = \Phi_6(t) = t^2 - t + 1$$

and $\Phi_6(1) = 1$ makes Φ_6 a unit. So genuine K-theory tells the two orbifolds apart, while integral cohomology and Borel K-theory do not.

In Proposition 1.4.3 the roots of $\prod_i (x + a_i)$ are the restrictions $-a_i$ of x to the fixed points. To see why the fixed points carry so much, we turn to differential forms.

1.5 The Cartan model and localisation

The whole section rests on the following basic fact.

Proposition 1.5.1 ([23]). *Let $\pi\colon P \to B$ be a principal G-bundle with G connected, and for $\xi \in \mathfrak{g}$ let ξ_P be the vector field generated by the action. Then π^* is an isomorphism*

$$\Omega^*(B) \xrightarrow{\cong} \Omega^*(P)_{\mathrm{bas}} = \Big\{\alpha \in \Omega^*(P) \;\Big|\; \iota_{\xi_P}\alpha = 0,\ L_{\xi_P}\alpha = 0 \text{ for all } \xi \in \mathfrak{g}\Big\}.$$

That is, a form on B is the same as a form on P that is horizontal, killed by the vertical directions, and invariant along them.

Now take the principal G-bundle $EG \times M \to EG \times_G M$. By Proposition 1.5.1, forms on $EG \times_G M$ are the basic forms on $EG \times M$, so to describe $\Omega^*(EG \times_G M)$ we need at least $\Omega^*(EG)$. As EG is infinite-dimensional, we replace $\Omega^*(EG)$ by the following algebraic model.

The point is that $EG \to BG$ is a principal bundle, and a connection $A \in \Omega^1(EG; \mathfrak{g})$ and its curvature F satisfy

$$dA = F - \tfrac{1}{2}[A, A], \qquad dF = -[A, F],$$

the second being the Bianchi identity. The Weil algebra has one generator mimicking A, one mimicking F, and these two equations as its differential. This is enough, since the connection and its curvature are the two fundamental forms of gauge theory.

Definition 1.5.2. Let $e_1, \dots, e_r$ be a basis of $\mathfrak{g}$ with dual basis $e^1, \dots, e^r$. The *Weil algebra* is $W(\mathfrak{g}) = \Lambda(\mathfrak{g}^*) \otimes S(\mathfrak{g}^*)$, with generators $\theta^a = e^a \otimes 1$ of degree 1 and $u^a = 1 \otimes e^a$ of degree 2. Writing $\theta = \sum_a \theta^a e_a$ and $u = \sum_a u^a e_a$, its differential is

$$d_W \theta = u - \tfrac{1}{2}[\theta, \theta], \qquad d_W u = -[\theta, u].$$

So θ is a universal connection and u its curvature. For a torus the brackets vanish, $d_W \theta^a = u^a$ and $d_W u^a = 0$, so for example $u^k = d_W(\theta u^{k-1})$ for a circle. In general $H^*(W(\mathfrak{g})) = \mathbb{R}$, so $W(\mathfrak{g})$ models the contractible space EG. Contraction with $\xi \in \mathfrak{g}$ acts as on a connection, $\iota_\xi \theta = \xi$ and $\iota_\xi u = 0$, and $L_\xi = d_W \iota_\xi + \iota_\xi d_W$.

The group acts diagonally on $EG \times M$, so contraction and Lie derivative act on both factors. Replacing $\Omega^*(EG)$ by $W(\mathfrak{g})$ gives the following.

Definition 1.5.3. Let M be a G-manifold. On $W(\mathfrak{g}) \otimes \Omega^*(M)$ put the differential $D = d_W \otimes 1 + 1 \otimes d$, and for $\xi \in \mathfrak{g}$

$$\iota_\xi^{\mathrm{tot}} = \iota_\xi \otimes 1 + 1 \otimes \iota_{\xi_M}, \qquad L_\xi^{\mathrm{tot}} = L_\xi \otimes 1 + 1 \otimes L_{\xi_M},$$

with the usual signs. The *Weil model* of M is the subcomplex of *basic* elements, those killed by ι_ξ^{tot} and L_ξ^{tot} for all $\xi \in \mathfrak{g}$.

By Proposition 1.5.1, this is the natural model for $H^*_G(M; \mathbb{R})$.

Example 1.5.4. Let $G = S^1$, so $W(\mathfrak{g}) = \mathbb{R}[u] \otimes \Lambda(\theta)$ with $d_W \theta = u$, and let X be the vector field of the generator e, so $\iota_e \theta = 1$ and $\iota_e u = 0$. Every element of $W(\mathfrak{g}) \otimes \Omega^*(M)$ is uniquely $A + \theta B$ with $A, B \in \mathbb{R}[u] \otimes \Omega^*(M)$, and

$$\iota^{\mathrm{tot}}(A + \theta B) = \iota_X A + (\iota_e \theta) B - \theta\, \iota_X B = B + \iota_X A - \theta\, \iota_X B.$$

So $A+\theta B$ is horizontal exactly when $B=-\iota_X A$. The horizontal condition forces the θ-part, and

$$\Phi\colon \mathbb{R}[u]\otimes\Omega^*(M)\longrightarrow \big(W(\mathfrak{g})\otimes\Omega^*(M)\big)_{\mathrm{hor}},\qquad A\longmapsto A-\theta\,\iota_X A,$$

is a bijection. As S^1 is abelian, $L_e\theta = L_e u = 0$, so $L^{\mathrm{tot}}\Phi(A)=\Phi(L_X A)$, and $\Phi(A)$ is basic exactly when A is invariant. For invariant A, Cartan's formula gives $d\iota_X A=-\iota_X dA$, so

$$D\Phi(A)=dA-u\,\iota_X A+\theta\,d\iota_X A=\Phi(dA-u\,\iota_X A).$$

So the Weil model of M is $\Omega^*(M)^{S^1}[u]$ with differential $d-u\iota_X$.

Definition 1.5.5. The *Cartan model* of a G-manifold M is $\Omega_G(M)=\big(S(\mathfrak{g}^*)\otimes\Omega^*(M)\big)^G$, the G-invariant polynomial maps $\alpha\colon\mathfrak{g}\to\Omega^*(M)$, with $\mathfrak{g}^*$ in degree 2 and differential

$$(d_G\alpha)(\xi)=d\big(\alpha(\xi)\big)-\iota_{\xi_M}\big(\alpha(\xi)\big).$$

For a circle, $\Omega_{S^1}(M)=\Omega^*(M)^{S^1}[u]$ and $d_G=d-u\iota_X$.

By Cartan's formula $(d_G^2\alpha)(\xi)=-L_{\xi_M}\alpha(\xi)=0$ for invariant α. The computation above extends to compact G, and the Weil and Cartan models are isomorphic [23].

Theorem 1.5.6 (Equivariant de Rham theorem [23])**.** *For a compact connected Lie group G acting on a manifold M,*

$$H\big(\Omega_G(M),d_G\big)\cong H^*_G(M;\mathbb{R})$$

as algebras over $S(\mathfrak{g}^*)^G=H^*(BG;\mathbb{R})$.

Remark 1.5.7. For a point, $\Omega_G(\mathrm{pt})=S(\mathfrak{g}^*)^G$ with $d_G=0$, so Theorem 1.5.6 gives $H^*(BG;\mathbb{R})\cong S(\mathfrak{g}^*)^G$. This is the Chern–Weil isomorphism.

Example 1.5.8. The Cartan class u is the class u of Example 1.3.10. On $\mathbb{C}_w$ with $z=|z|e^{i\varphi}$ we have $X=2\pi w\,\partial_\varphi$. For a function $h(|z|)$ equal to -1 near 0 and to 0 for $|z|\geq 1$, the form

$$\mathrm{Th}=d_G\Big(h\,\frac{d\varphi}{2\pi}\Big)=dh\wedge\frac{d\varphi}{2\pi}-wu\,h$$

is smooth, compactly supported, d_G-closed and integrates to 1 on $\mathbb{C}_w$, so it is a Thom form. Its restriction to the origin is wu, which is $e_{S^1}(\mathbb{C}_w)$ by Example 1.3.10. With $d+u\iota_X$ in place of d_G one would get $-wu$.

Proposition 1.5.9 (Equivariant Stokes)**.** *Let M be compact and oriented of dimension m, with boundary ∂M. Then*

$$\int_M d_G\alpha = \int_{\partial M} \alpha$$

for every $\alpha \in \Omega_G(M)$. In particular, if $\partial M = \varnothing$, integration induces a map $\int_M \colon H^*_G(M;\mathbb{R}) \to H^{*-m}_G(\mathrm{pt};\mathbb{R})$. The same holds for compactly supported forms on a manifold without boundary.

Proof. For $\xi \in \mathfrak{g}$,

$$\int_M (d_G\alpha)(\xi) = \int_M d\Big(\alpha(\xi)\Big) - \int_M \iota_{\xi_M}\Big(\alpha(\xi)\Big) = \int_{\partial M} \alpha(\xi) - 0,$$

by Stokes for the first term. The second vanishes, since ι_{ξ_M} lowers degree by one, so $\iota_{\xi_M}\Big(\alpha(\xi)\Big)$ has no part of degree m. □

This is the key point. By Proposition 1.5.9, on a closed M we may add any d_G-exact form to α before integrating, and the integral does not change. For example, let β be an invariant 1-form. Then $d_G(d_G\beta) = 0$, so

$$e^{t\,d_G\beta} - 1 = \sum_{k\geq 1} \frac{t^k}{k!}(d_G\beta)^k = d_G\eta_t, \qquad \eta_t = \beta \sum_{k\geq 1} \frac{t^k}{k!}(d_G\beta)^{k-1},$$

and for every d_G-closed α

$$\begin{aligned} \int_M \alpha\, e^{t\,d_G\beta} &= \int_M \alpha + \int_M \alpha\left(e^{t\,d_G\beta} - 1\right) \\ &= \int_M \alpha + (-1)^{|\alpha|} \int_M d_G(\alpha\,\eta_t) = \int_M \alpha. \end{aligned}$$

Here $e^{t\,d_G\beta}$ is a power series in u, and Proposition 1.5.9 applies to each coefficient.

Choosing β well, the integral localises to the fixed points.

Theorem 1.5.10 (Atiyah–Bott, Berline–Vergne [6, 11])**.** *Let S^1 act on a compact oriented manifold M of dimension $2n$ with isolated fixed points. At each fixed point p identify $T_pM \cong \bigoplus_{j=1}^n \mathbb{C}_{w_{p,j}}$, compatibly with the orientation. Then for every d_G-closed $\alpha \in \Omega_{S^1}(M)$,*

$$\int_M \alpha = \sum_{p \in M^{S^1}} \frac{\alpha|_p}{e_{S^1}(T_pM)} = \sum_{p \in M^{S^1}} \frac{\alpha|_p}{\Big(\prod_j w_{p,j}\Big)u^n} \qquad \text{in } \mathbb{R}[u,u^{-1}].$$

Sketch. Choose an invariant metric g which is standard in linear coordinates $z_j = r_j e^{i\varphi_j}$ near each fixed point, and put $\beta = g(X,\cdot)$. Then $d_G\beta = d\beta - u|X|^2$, and by the deformation above, for $s \geq 0$,

$$\int_M \alpha = \int_M \alpha\, e^{-su|X|^2} e^{s\,d\beta}.$$

Take $u > 0$ and let $s \to \infty$, see Figure 1.12. Since $d\beta$ is a 2-form, $e^{s\,d\beta}$ is a polynomial in s of degree at most n. Away from the fixed points $|X|^2 \geq c > 0$, so there the integrand is at most a polynomial in s times e^{-scu}, which tends to 0. Near p we have $X = \sum_j 2\pi w_j \partial_{\varphi_j}$, so

$$|X|^2 = \sum_j 4\pi^2 w_j^2 |z_j|^2, \qquad d\beta = \sum_j 4\pi w_j \, dx_j \wedge dy_j.$$

Substitute $z = z'/\sqrt{s}$. The Gaussian factor no longer depends on s, each $dx_j \wedge dy_j$ gains a factor $1/s$, and α is evaluated at points tending to p. The top-degree part $s^n \prod_j 4\pi w_j \, dx_j \, dy_j$ of $e^{s\,d\beta}$ then gives a limit independent of s, while every term in which α contributes a form of positive degree carries fewer powers of s and tends to 0. So the contribution of p is

$$\alpha|_p \prod_j \int_{\mathbb{R}^2} 4\pi w_j \, e^{-4\pi^2 w_j^2 u |z_j'|^2} \, dx_j' \, dy_j' = \alpha|_p \prod_j \frac{1}{w_j u}.$$

Both sides are Laurent polynomials in u that agree for $u > 0$. □

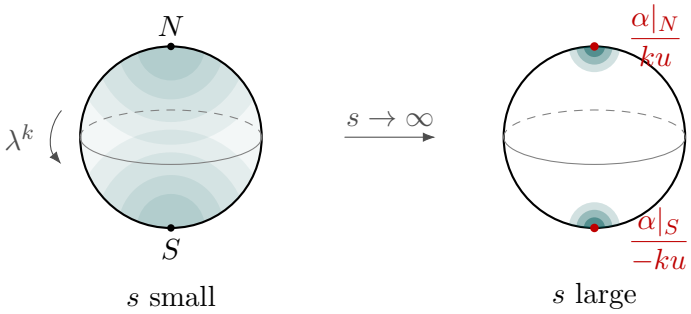


Figure 1.12: As s grows, $e^{-su|X|^2}$ concentrates at the fixed points.

The same argument works for a torus.

Theorem 1.5.11. *Let a torus T act on a compact oriented manifold M, and let F run over the components of M^T, with inclusions i_F and normal bundles ν_F, oriented compatibly with M and F. Then for every d_T-closed $\alpha \in \Omega_T(M)$,*

$$\int_M \alpha = \sum_F \int_F \frac{i_F^* \alpha}{e_T(\nu_F)},$$

where the right-hand side is computed after inverting the non-zero weights.

If $F = \{p\}$ is a point, $e_T(\nu_F)$ is the product of the weights of T_pM, as in (1.3.1). In general it is this product plus nilpotent terms from the Chern classes of ν_F, so it

is invertible once the weights are inverted. For the proof, restrict to a circle $S^1 \subset T$ with the same fixed points, which exists since only finitely many stabilisers occur, and apply the circle case of [6], where F need not be a point. As the directions of such circles are dense, the formula holds for T.

Example 1.5.12. Let S^1 act on $\mathbb{CP}^2$ by $\lambda \cdot [z_0 : z_1 : z_2] = [z_0 : z_1 : \lambda z_2]$. A point $[z]$ is fixed exactly when $z_2 = 0$ or $z_0 = z_1 = 0$, so M^{S^1} is the line $F = \{z_2 = 0\} \cong \mathbb{CP}^1$, on which S^1 acts trivially, together with the point P_2. We compute the two contributions.

On F, the fibre $\mathbb{C}z$ of $\mathcal{O}(-1)$ has weight 0, so $\mathcal{O}(1)|_F$ is the line bundle $\mathcal{O}(1)$ of $\mathbb{CP}^1$ with trivial action, and $i_F^* x = h$, the hyperplane class, with $h^2 = 0$. The normal direction to F is the z_2-direction, so $\nu_F = \mathrm{Hom}(\mathcal{O}(-1), \mathbb{C}_1) = \mathcal{O}(1) \otimes \mathbb{C}_1$ and $e_{S^1}(\nu_F) = h + u$. As $h^2 = 0$,

$$\frac{1}{e_{S^1}(\nu_F)} = \frac{1}{u} - \frac{h}{u^2},$$

and $\int_F$ picks out the coefficient of h.

At P_2 we have $x|_{P_2} = -u$ by Proposition 1.4.3(d), and the tangent weights are $0 - 1 = -1$ twice, so $e_{S^1}(T_{P_2}\mathbb{CP}^2) = u^2$. Theorem 1.5.11 gives

$$\int_{\mathbb{CP}^2} x = \int_F h\Big(\frac{1}{u} - \frac{h}{u^2}\Big) + \frac{-u}{u^2} = \frac{1}{u} - \frac{1}{u} = 0, \qquad \int_{\mathbb{CP}^2} x^2 = \int_F h^2\Big(\frac{1}{u} - \frac{h}{u^2}\Big) + \frac{u^2}{u^2} = 1,$$

as expected.

As an application, localisation gives intersection numbers of non-compact cycles, as long as the fixed points are compact. If M is not compact but M^T is, we define $\int_M \alpha$ by the right-hand side of Theorem 1.5.11. For $\mathbb{C}_w$ this gives $\int_{\mathbb{C}_w} 1 = 1/(wu)$, the value of the Gaussian integral in the proof of Theorem 1.5.10.

Example 1.5.13. Let $T = (S^1)^2$ act on $\mathbb{C}^2$ by $(t_1, t_2) \cdot (z_1, z_2) = (t_1 z_1, t_2 z_2)$, with $H_T^*(\mathrm{pt}) = \mathbb{R}[u_1, u_2]$. The only fixed point is 0, with $e_T(T_0\mathbb{C}^2) = u_1 u_2$, so $\int_{\mathbb{C}^2} \alpha = \alpha|_0/(u_1 u_2)$. By Corollary 1.3.8 the axes $Z_1 = \{z_2 = 0\}$ and $Z_2 = \{z_1 = 0\}$ have classes $[Z_1]_T = u_2$ and $[Z_2]_T = u_1$. So

$$\int_{\mathbb{C}^2} [Z_1]_T [Z_2]_T = \frac{u_1 u_2}{u_1 u_2} = 1, \qquad \int_{\mathbb{C}^2} [Z_1]_T^2 = \frac{u_2^2}{u_1 u_2} = \frac{u_2}{u_1}.$$

The two axes meet once, transversally. Without the action Z_1 could be pushed off itself to $\{z_2 = c\}$, but this is not invariant, and the equivariant self-intersection is not a number. It remembers the weights.

1.6 Fixed points and GKM theory

In the previous section the fixed points determined every integral. In fact much more is true, and the fixed points control the whole equivariant cohomology.

Symplectic geometry is the natural bridge, and we start by looking at the moment map in the Cartan model.

Definition 1.6.1. Let a compact Lie group G act on a symplectic manifold (M, ω) preserving ω. A *moment map* is a G-equivariant map $\mu\colon M \to \mathfrak{g}^*$, with G acting on $\mathfrak{g}^*$ by the coadjoint action, such that $\iota_{\xi_M}\omega = d\mu^\xi$ for all $\xi \in \mathfrak{g}$, where $\mu^\xi = \langle \mu, \xi \rangle$. The action is *Hamiltonian* if a moment map exists.

Lemma 1.6.2. *For a G-equivariant map $\mu\colon M \to \mathfrak{g}^*$, the equivariant form $\omega + \mu\colon \xi \mapsto \omega + \mu^\xi$ is d_G-closed exactly when μ is a moment map.*

Proof. Since $d\omega = 0$ and ι_{ξ_M} kills functions, $(d_G(\omega + \mu))(\xi) = d\mu^\xi - \iota_{\xi_M}\omega$. □

So a moment map is exactly an equivariant extension of ω, and $[\omega + \mu] \in H^2_G(M)$ is the equivariant symplectic class.

Being Hamiltonian is a real condition.

Example 1.6.3. Let S^1 act on the torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ by rotating the first factor, with $\omega = d\theta_1 \wedge d\theta_2$. Then $X = \partial_{\theta_1}$, and $\iota_X\omega = d\theta_2$ is closed, so $L_X\omega = d\iota_X\omega = 0$ and the action preserves ω. Suppose μ were a moment map. Then $d\mu = \iota_X\omega = d\theta_2$. Integrating along the second circle $\gamma(s) = (0, s)$, $0 \le s \le 1$, gives

$$0 = \mu\big(\gamma(1)\big) - \mu\big(\gamma(0)\big) = \int_\gamma d\mu = \int_\gamma d\theta_2 = 1,$$

a contradiction. So the action has no moment map.

The real reason is that a maximum of μ would be a fixed point.

Lemma 1.6.4. *Let μ be a moment map and $\xi \in \mathfrak{g}$. The critical points of μ^ξ are exactly the zeros of ξ_M.*

Proof. We have $d\mu^\xi = \iota_{\xi_M}\omega$, and ω is non-degenerate. □

So for generic ξ the fixed points are the critical points of μ^ξ, and as in Morse theory we expect them to control the topology. We first prove a weaker but general result, which needs no moment map. Let $R = H^*(BT; \mathbb{Q})$ and $S = R \setminus \{0\}$. We call an R-module *torsion* if each of its elements is killed by some $s \in S$. Localising at S kills exactly the torsion. We start with one orbit.

Lemma 1.6.5. *Let $H \subsetneq T$ be a closed subgroup. Then $H_T^*(T/H;\mathbb{Q})$ is a torsion R-module.*

Proof. We have $ET \times_T (T/H) \cong ET/H \simeq BH$, so $H_T^*(T/H;\mathbb{Q}) \cong H^*(BH;\mathbb{Q})$, with R acting through restriction. As T/H is a torus of positive dimension, it has a non-trivial character, and pulling it back gives a non-trivial character χ of T with $\chi|_H = 1$. Then $\alpha = e_T(\mathbb{C}_\chi)$ is non-zero in R, and it restricts to $e_H(\mathbb{C}_{\chi|_H}) = 0$ in $H^*(BH;\mathbb{Q})$, so α kills $H_T^*(T/H;\mathbb{Q})$. □

A cell only shifts degrees.

Lemma 1.6.6. *If Y is obtained from a T-space A by attaching finitely many cells $T/H \times D^n$ with $H \neq T$, then $H_T^*(Y, A;\mathbb{Q})$ is torsion.*

Proof. By excision and the long exact sequence of a triple it suffices to treat one cell. As T acts trivially on D^n, $H_T^*\big(T/H \times (D^n, S^{n-1});\mathbb{Q}\big) \cong H_T^{*-n}(T/H;\mathbb{Q})$, which is torsion by Lemma 1.6.5. □

Theorem 1.6.7 (Borel localisation [13])**.** *Let a torus T act on a compact manifold, or more generally on a finite T-CW complex X, and let $i\colon X^T \hookrightarrow X$ be the inclusion of the fixed points. Then*

$$S^{-1}i^*\colon S^{-1}H_T^*(X;\mathbb{Q}) \longrightarrow S^{-1}H_T^*(X^T;\mathbb{Q}) = S^{-1}R \otimes H^*(X^T;\mathbb{Q})$$

is an isomorphism. Equivalently, the kernel and the cokernel of i^* are torsion.

Proof. The fixed points form a subcomplex, and attaching the remaining cells one at a time gives a filtration $X^T = X_0 \subset X_1 \subset \cdots \subset X_N = X$ as in Lemma 1.6.6. Consider the long exact sequence of the triple $X^T \subset X_{j-1} \subset X_j$,

$$\cdots \to H_T^*(X_j, X_{j-1}) \to H_T^*(X_j, X^T) \to H_T^*(X_{j-1}, X^T) \to \cdots .$$

Its first term is torsion by Lemma 1.6.6, and the third by induction on j. Then so is the middle one. If $s \in S$ kills the image of b in the third term, then sb comes from the first, which some $t \in S$ kills, so $tsb = 0$. Hence $H_T^*(X, X^T;\mathbb{Q})$ is torsion. Localisation is exact and kills torsion, so localising the long exact sequence of the pair (X, X^T) shows that $S^{-1}i^*$ is an isomorphism. In the same sequence, $\ker i^*$ is a quotient and $\operatorname{coker} i^*$ a submodule of torsion modules. □

In terms of $\operatorname{Spec} R$, write $A = H_T^*(X;\mathbb{Q})$ and $B = H_T^*(X^T;\mathbb{Q})$. They are R-modules, so they give quasi-coherent sheaves $\widetilde{A}$ and $\widetilde{B}$ on $\operatorname{Spec} R$, which becomes $\mathfrak{t}_{\mathbb{C}}$ after tensoring with $\mathbb{C}$, and i^* gives a map $\widetilde{i^*}\colon \widetilde{A} \to \widetilde{B}$.

Proposition 1.6.8. *Let $\eta = (0)$ be the generic point of* $\operatorname{Spec} R$*, and let* $K = \operatorname{Frac} R = S^{-1}R$*. Then* $\widetilde{A}_\eta = A \otimes_R K$ *and* $\widetilde{B}_\eta = B \otimes_R K$*, so Theorem 1.6.7 says exactly that* $\widetilde{i^*}$ *is an isomorphism at the generic point.*

Proof. The stalk at the prime (0) is the localisation at $R \setminus (0) = S$. □

Equivalently, $\ker i^*$ and $\operatorname{coker} i^*$ vanish at the generic point, that is, they are torsion. Geometrically, an element n of an R-module with $fn = 0$ for some $f \neq 0$ vanishes on the open set $\{f \neq 0\}$, where f is invertible, so it lives on $\{f = 0\}$. So the difference between A and B lives on proper closed subsets of $\operatorname{Spec} R$.

The proof shows which subsets. An orbit T/H contributes $H^*(BH;\mathbb{Q}) = \operatorname{Sym}(\mathfrak{h}^*_\mathbb{Q})$, the coordinate ring of the subspace $\mathfrak{h}_\mathbb{Q} \subset \mathfrak{t}_\mathbb{Q}$, so it lives over $\mathfrak{h}$. Fixed points live over all of $\mathfrak{t}$, and the other orbits over the finitely many proper subspaces $\mathfrak{h}$ that occur as Lie algebras of stabilisers. For instance, if $\mathfrak{h} = \ker \alpha$ is a hyperplane, then $H^*(BH;\mathbb{Q}) = R/(\alpha)$ lives over $\{\alpha = 0\}$. We call these hyperplanes *walls*. They come from the orbits of dimension one, which will reappear below.

So the proof gives more than Theorem 1.6.7, since it suffices to invert a single element. For each stabiliser $H \neq T$ choose a non-zero $\alpha_H \in R$ of degree two vanishing on $\mathfrak{h}$, as in Lemma 1.6.5, and put $\Delta = \prod_H \alpha_H$. Then the relative term is killed by a power of Δ, and

$$\widetilde{A}|_{D(\Delta)} \cong \widetilde{B}|_{D(\Delta)}, \qquad D(\Delta) = \{\Delta \neq 0\},$$

so away from finitely many hyperplanes the fixed points see everything, see Figure 1.13.

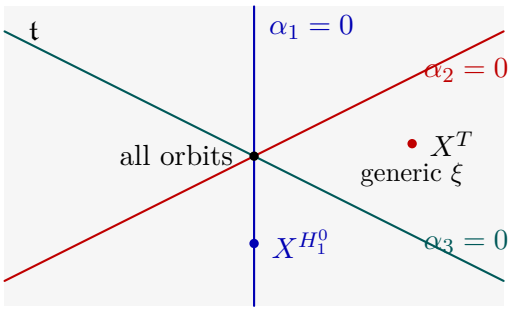


Figure 1.13: The walls $\alpha_i = 0$. Away from them, on $D(\Delta)$, only X^T survives.

Remark 1.6.9. If M^T is finite, $\widetilde{B}$ is one sheet $\mathcal{O}_{\operatorname{Spec} R}$ for each fixed point, so a class on M^T is a tuple of polynomials $(f_p)_{p \in M^T}$. The sheets are independent, but classes coming from M are sewn together along the walls. If p and q are joined by an invariant sphere of weight α, this sphere is seen on the wall $\alpha = 0$, and it forces f_p and f_q to agree there, that is, $f_p - f_q \in (\alpha)$.

For a Hamiltonian action we get more, since a generic component μ^ξ of the moment map is a Morse–Bott function with critical set M^T, and standard Morse theory then describes $H_T^*(M;\mathbb{Q})$ as an R-module.

Theorem 1.6.10 (Kirwan injectivity [44])**.** *Let T act on a compact symplectic manifold M with moment map μ. Then $H_T^*(M;\mathbb{Q})$ is a free R-module of rank $\dim H^*(M^T;\mathbb{Q})$, and $i^*\colon H_T^*(M;\mathbb{Q}) \to H_T^*(M^T;\mathbb{Q})$ is injective.*

We state without proof a slightly stronger result, which leads to the theory of Goresky, Kottwitz and MacPherson [21], GKM theory for short. It says that the image of i^* is cut out by the one-skeleton alone.

Theorem 1.6.11 (Chang–Skjelbred [18])**.** *Let T act on a compact manifold M with $H_T^*(M;\mathbb{Q})$ free over R, and let $M_1 = \{x \in M : \dim T\cdot x \le 1\}$ be the one-skeleton. Then*

$$0 \to H_T^*(M;\mathbb{Q}) \xrightarrow{i^*} H_T^*(M^T;\mathbb{Q}) \xrightarrow{\delta} H_T^{*+1}(M_1, M^T;\mathbb{Q})$$

is exact. So a class on M^T comes from M exactly when it extends to M_1.

By Theorem 1.6.10 it applies to every Hamiltonian action. Its meaning becomes clear from the long exact sequence of the pair (M_1, M^T), in which $\ker\delta$ is the image of $H_T^*(M_1;\mathbb{Q})$. So the theorem says

$$\operatorname{Im}\Big(H_T^*(M;\mathbb{Q}) \to H_T^*(M^T;\mathbb{Q})\Big) = \operatorname{Im}\Big(H_T^*(M_1;\mathbb{Q}) \to H_T^*(M^T;\mathbb{Q})\Big).$$

Thus the compatibility conditions satisfied by global classes already come from the one-skeleton M_1. In terms of $\operatorname{Spec} R$, the fixed point data are controlled by the walls, the codimension-one part of Figure 1.13.

The next question is when M_1 is simple enough to be recorded combinatorially. This leads to GKM manifolds. Goresky, Kottwitz and MacPherson showed that for a well-behaved class of torus actions the equivariant cohomology can be read off from a labelled graph. Conventions vary in the literature, and we only need the version with isolated fixed points in the Hamiltonian setting.

Let (M,ω) be a compact symplectic manifold with a Hamiltonian T-action and moment map $\mu\colon M \to \mathfrak{t}^*$. We call it a *compact Hamiltonian T-manifold*. Choose a T-invariant ω-compatible almost complex structure J. At each fixed point $p \in M^T$ the tangent space T_pM is then a complex representation of T, and it splits into weight lines,

$$T_pM \cong \bigoplus_{j=1}^{n} \mathbb{C}_{\alpha_{p,j}}. \tag{1.6.1}$$

Here $\alpha_{p,j}$ is a character of T, and we also write $\alpha_{p,j}$ for its Euler class $e_T(\mathbb{C}_{\alpha_{p,j}}) \in H^2(BT;\mathbb{Q})$, as in Section 1.3.

Definition 1.6.12 (GKM manifold)**.** A compact oriented T-manifold M is a *GKM manifold* over $\mathbb{Q}$ if

1. the action is equivariantly formal over $\mathbb{Q}$, that is, $H_T^*(M;\mathbb{Q})$ is a free R-module;
2. the fixed point set M^T is finite;
3. at every $p \in M^T$ the weights of T_pM are pairwise linearly independent.

Remark 1.6.13. A common equivalent form of condition (1) is that the Serre spectral sequence of $M \to ET \times_T M \to BT$ collapses at E_2, so $H_T^*(M;\mathbb{Q}) \cong R \otimes H^*(M;\mathbb{Q})$ as R-modules.

For a general T-manifold the weights at an isolated fixed point are only defined up to sign, which does not affect condition (3).

Proposition 1.6.14. *Let M be a compact Hamiltonian T-manifold with M^T finite and with pairwise linearly independent weights at every fixed point. Then M is a GKM manifold.*

Proof. Condition (1) is Theorem 1.6.10, and the other two are the hypotheses. □

Pairwise independence makes the one-skeleton look like a graph. Near a fixed point p the action is the linear action on $T_pM \cong \bigoplus_j \mathbb{C}_{\alpha_{p,j}}$. A point with non-zero coordinates in two weight directions has an orbit of dimension at least two, as the two weights are linearly independent. So near p the one-skeleton is the union of the weight axes.

The following property is the most important part of GKM theory.

Proposition 1.6.15. *Let M be a compact Hamiltonian GKM manifold. Then*

$$M_1 = M^T \cup \bigcup_{e \in E} S_e$$

for a finite set E of T-invariant two-spheres S_e, each containing exactly two fixed points, and the spheres through a fixed point p correspond to the weights at p.

Proof. Let $x \in M_1 \setminus M^T$. Its stabiliser has identity component a codimension-one subtorus H, and let C be the component of M^H containing x. The circle T/H acts on C in a Hamiltonian way, so the maximum of its moment map is a fixed point $p \in C$. Then $T_pC = (T_pM)^H$ is a single weight axis, since by pairwise independence only one weight at p vanishes on $\mathrm{Lie}(H)$. So C is a compact connected surface with a Hamiltonian circle action, hence a sphere whose two fixed points are the extrema of the moment map. There are finitely many such spheres, since each fixed point has finitely many weight axes. □

Figure 1.14 shows the picture.

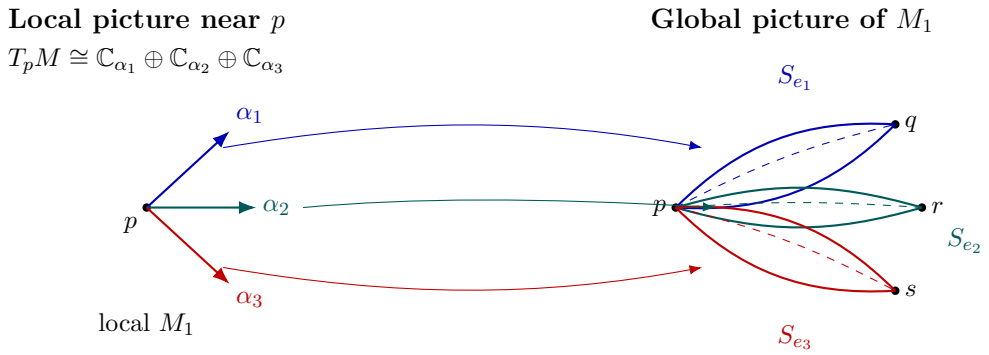


Figure 1.14: The weight axes at p close up to the spheres of M_1.

We encode the one-skeleton by a labelled graph.

Definition 1.6.16 (GKM graph)**.** Let M be a compact Hamiltonian GKM manifold. Its *GKM graph* $\Gamma_M = (V, E, \alpha)$ has vertex set $V = M^T$ and an edge e joining p and q for each sphere S_e with fixed points p and q. The edge is labelled by the weight α_e of $T_pS_e \cong \mathbb{C}_{\alpha_e}$. At q the weight is $-\alpha_e$, so the label of an unoriented edge is defined up to sign.

Each edge gives one relation. The subgroup $\ker \alpha_e$ acts trivially on S_e, so T acts on S_e through the circle $T/\ker \alpha_e \cong S^1$, given by α_e, which rotates S_e with weights ± 1. By Example 1.1.9 with $k = 1$, pulled back along $BT \to BS^1$, which sends u to $e_T(\mathbb{C}_{\alpha_e}) = \alpha_e$,

$$\operatorname{Im}\Big(H_T^*(S_e;\mathbb{Q}) \to H_T^*(p) \oplus H_T^*(q)\Big) = \{(f_p, f_q) \in R \oplus R : \ f_p - f_q \in (\alpha_e)\}.$$

That is, f_p and f_q agree on the wall $\alpha_e = 0$, as in Remark 1.6.9.

Definition 1.6.17 (GKM graph algebra)**.** Let $\Gamma_M = (V, E, \alpha)$ be the GKM graph of M. Its *graph algebra* is

$$H^*(\Gamma_M, \alpha) = \Big\{ f : V \to R \;\Big|\; f(p) - f(q) \in (\alpha_e) \text{ for every edge } e = (p, q) \Big\},$$

with pointwise multiplication.

Theorem 1.6.18 (GKM theorem [21])**.** *Let M be a compact Hamiltonian GKM manifold. Then restriction to M^T identifies $H_T^*(M;\mathbb{Q})$ with $H^*(\Gamma_M, \alpha)$.*

Proof. By Theorem 1.6.10, restriction to M^T is injective, so $H_T^*(M;\mathbb{Q})$ is a subalgebra of $H_T^*(M^T;\mathbb{Q}) = \bigoplus_{p\in M^T} R$. By Theorem 1.6.11 its image is the image of $H_T^*(M_1;\mathbb{Q})$. By Proposition 1.6.15, M_1 is the union of the spheres S_e, which meet only at fixed points, so a tuple (f_p) extends to M_1 exactly when it extends over every S_e, that is, when $f_p - f_q \in (\alpha_e)$ for every edge. □

Remark 1.6.19. This makes Remark 1.6.9 precise. The sheets are sewn along the walls $\alpha_e = 0$, one for each edge, and the edge labels are the equivariant Euler classes of the tangent lines T_pS_e.

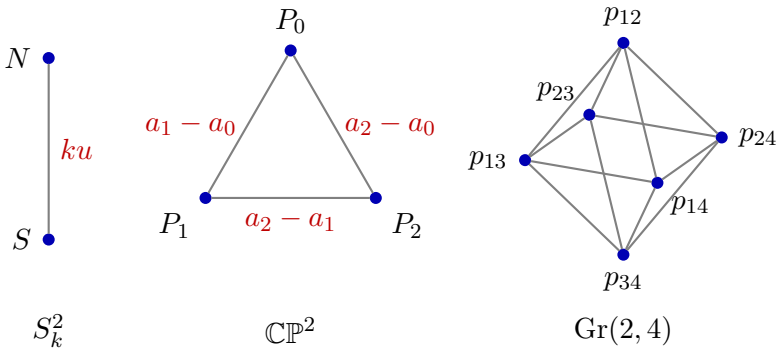


Figure 1.15: Three GKM graphs.

Example 1.6.20. Let S^1 rotate S^2 with weight $k \neq 0$, as in Example 1.1.9. The fixed points are the poles, joined by one edge, the sphere itself, with label ku (Figure 1.15). So Theorem 1.6.18 gives

$$H_{S^1}^*(S^2;\mathbb{Q}) = \{(f_N, f_S) \in \mathbb{Q}[u]^2 : \ f_N - f_S \in (ku)\},$$

the rational form of the ring computed in Example 1.1.9 by Mayer–Vietoris, where the same description holds over $\mathbb{Z}$.

Example 1.6.21. Let $T = (S^1)^{n+1}$ act on $\mathbb{CP}^n$ by $(\lambda_0,\dots,\lambda_n)\cdot[z_0 : \cdots : z_n] = [\lambda_0 z_0 : \cdots : \lambda_n z_n]$, as in Proposition 1.4.3. A point is fixed exactly when only one coordinate is non-zero, so the fixed points are $P_i = [0 : \cdots : 1 : \cdots : 0], i = 0,\dots,n$. Near P_i we use the coordinates $w_j = z_j/z_i$, $j \neq i$, in which the action is $w_j \mapsto \lambda_j\lambda_i^{-1}w_j$. So the weights at P_i are $a_j - a_i$, $j \neq i$. They are pairwise independent, the sphere of $a_j - a_i$ is the line P_iP_j, and the GKM graph is the complete graph on $P_0,\dots,P_n$ with labels $a_j - a_i$ (Figure 1.15). The action preserves the Fubini–Study form and is Hamiltonian, so Theorem 1.6.18 gives

$$H_T^*(\mathbb{CP}^n;\mathbb{Q}) = \{(f_0,\dots,f_n) \in R^{n+1} \mid f_i - f_j \in (a_i - a_j)\},$$

and one can check directly that this agrees with Proposition 1.4.3(a).

Finally, we work out one example in detail. It shows step by step the general procedure for a GKM manifold.

Example 1.6.22. Let $\mathrm{Gr}(2,n)$ be the space of two-dimensional subspaces $V \subset \mathbb{C}^n$, a compact complex manifold of complex dimension $2(n-2)$.

Step 1. The Hamiltonian action. The *Plücker embedding*

$$\iota\colon \mathrm{Gr}(2,n) \hookrightarrow \mathbb{P}(\Lambda^2\mathbb{C}^n) \cong \mathbb{CP}^{\binom{n}{2}-1}, \qquad \langle v, w\rangle \longmapsto [v \wedge w],$$

is well defined because changing the basis v, w multiplies $v \wedge w$ by a non-zero determinant. Its coordinates π_{ij}, $i<j$, are the 2×2 minors of $(v\ w)$. Let $T = (S^1)^n$ act on $\mathbb{C}^n$ by $\lambda \cdot e_i = \lambda_i e_i$, with weights $a_1, \dots, a_n$. Then $e_i \wedge e_j$ has weight $a_i + a_j$, and ι is holomorphic and equivariant, so the Fubini–Study form and the moment map restrict to make $\mathrm{Gr}(2,n)$ a compact Hamiltonian T-manifold. From now on $n = 4$, so $\iota\colon \mathrm{Gr}(2,4) \hookrightarrow \mathbb{CP}^5$.

Step 2. The fixed points. The six weights $a_i + a_j$ are distinct, so the fixed points of $\mathbb{CP}^5$ are the points $[e_i \wedge e_j]$. They all lie on the image, so the fixed points of $\mathrm{Gr}(2,4)$ are the coordinate planes $p_{ij} = p_{ji} = \langle e_i, e_j\rangle$.

Step 3. The weights. Near p_{ij} every plane can be written uniquely as

$$V(z) = \Big\langle e_i + \sum_{m\notin\{i,j\}} z_{mi} e_m,\ e_j + \sum_{m\notin\{i,j\}} z_{mj} e_m \Big\rangle.$$

The four $z_{m\ell}$ are local coordinates. Applying λ and rescaling the two vectors by λ_i^{-1} and λ_j^{-1} gives $z_{m\ell} \mapsto (\lambda_m/\lambda_\ell) z_{m\ell}$. So the weights at p_{ij} are $a_m - a_\ell$ with $\ell \in \{i,j\}$ and $m \notin \{i,j\}$, for example $a_3 - a_1, a_4 - a_1, a_3 - a_2, a_4 - a_2$ at p_{12}. They involve different pairs of indices, so no two are proportional, and $\mathrm{Gr}(2,4)$ is GKM by Proposition 1.6.14.

Step 4. The spheres. For $m \notin \{i,j\}$, keeping e_j and moving the other vector inside $\langle e_i, e_m\rangle$ gives the sphere

$$S_{ij,jm} = \{\langle e_j, s e_i + t e_m\rangle \mid [s:t] \in \mathbb{CP}^1\}$$

from p_{ij} to p_{jm}. Near p_{ij} the coordinate t/s has weight $a_m - a_i$. The sphere is fixed by $H = \{\lambda_m = \lambda_i\}$, and at p_{ij} the H-fixed tangent space is just $\mathbb{C}_{a_m - a_i}$, so the sphere is the component of $\mathrm{Gr}(2,4)^H$ through p_{ij}. These spheres account for every weight at every fixed point, so by Proposition 1.6.15 they form the one-skeleton. Two vertices are joined exactly when they share one index, the edge from p_{ij} to p_{jm} has label $a_m - a_i$, and the graph is the octahedron of Figure 1.15.

Step 5. The ring. With $f_{ij} = f_{ji}$, Theorem 1.6.18 gives

$$H_T^*(\mathrm{Gr}(2,4);\mathbb{Q}) \cong \Big\{(f_{ij}) \in R^6 \;\Big|\; f_{ij} - f_{jm} \in (a_m - a_i) \text{ for distinct } i,j,m\Big\}.$$

Remark 1.6.23. Again, if one cares about intersection numbers, Theorem 1.5.11 computes them from the same data, $\int_{\mathrm{Gr}(2,4)} f = \sum_{i<j} f_{ij}/e_{ij}$, where e_{ij} is the product of the four weights at p_{ij}. We do not need them here.

We end this chapter by directing the graph.

Choose a generic $\xi \in \mathfrak{t}$. As in Morse theory we think of μ^ξ as a height function. On each edge, μ^ξ has its minimum at one end and its maximum at the other, so we orient the edge upwards, and the GKM graph becomes directed.

In fact, the Morse index of μ^ξ at p is twice the number of edges going down from p, so M has a cell structure with one $2k$-cell for each p with k edges going down. For $\mathrm{Gr}(2,4)$ this is Figure 1.16, which shows one cell in each of the dimensions $0, 2, 6, 8$ and two in dimension 4.

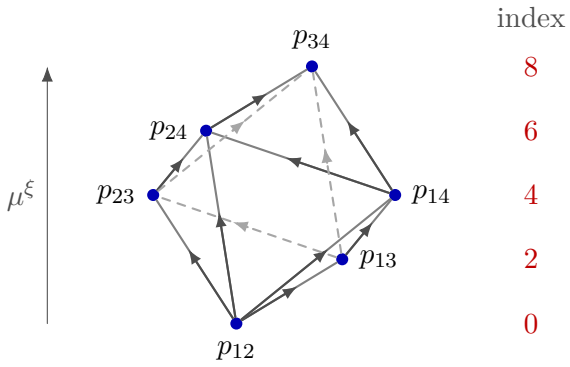


Figure 1.16: The GKM graph of $\mathrm{Gr}(2,4)$ directed by μ^ξ, with the Morse indices.

To sum up, this chapter rests on three ideas. First, removing an invariant submanifold costs its equivariant Euler class. Second, the fixed points control equivariant cohomology, through localisation, the Chang–Skjelbred sequence and the GKM theorem. Third, for Hamiltonian actions the moment map makes all of this work. So equivariant cohomology is a natural theory for symplectic geometry. In the next chapter we remove the unstable strata of $\|\mu\|^2$ one by one, and paying their Euler classes gives Kirwan surjectivity.

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